Department of
Aerospace Engineering
Faculty of Engineering
& Architectural Science

Semester (Term, Year)	Winter 2024
Course Code	AER 822
Course Section	1
Course Title	Avionics Design Project
Course Instructor	Dr. Guangjun Liu
Submission	Report
Submission Due Date	April 05, 2024
Title	Avionics Design Project CDR
Submission Date	April 05, 2024

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1. ABSTRACT

In this project, the Quanser 3DOF helicopter was analyzed, modelled, simulated, and controlled. In the modelling, it was found that through inspection, the Quanser helicopter was moving through three axes, the elevation, travel, and pitch axis. The transfer function of the elevation, pitch and travel axis were as seen in Section 4 for this report.

The controller was designed by testing the responsiveness of the system in various situations that could be tested in the laboratory. Initially, the system did not perform well and further tuning of the controllers using various methods was done. The controllers were updated such that the helicopter could handle certain maneuverability. After that, the focus was on increasing the responsiveness, accuracy and maneuverability of the helicopter.

A graphical user interface was also implemented to make it easier for the user to control the helicopter without going into the code. Trajectory and planning control was added in this new model which was used to define the pathway the helicopter took to get to a given destination in a specific length of time as per the first part of the final mission profile. The helicopter was able to follow a given path and travel 120 degrees, then 90 degrees back and 90 degrees forward again moved 15 degrees back and 15 degrees forward again before it landed.

A camera and a drill were also implemented as per the second part of the final mission profile. The camera was used to differentiate between the red blobs and the black blobs. Using the camera, the red blob angle was sent to the helicopter which was then used to get the helicopter to go to the board for drilling. The IRR values of the drill made it possible, using a function, to get the helicopter to the right place for the drill to start drilling. The addition of the drill made it possible for additional tuning of the travels PID values, though small tuning but it was significant. The project was a success in satisfying the two parts of the final mission profile.

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3. INTRODUCTION

The critical design review (CDR) serves as a pivotal checkpoint in the project life cycle, especially within engineering and design endeavours. It acts as a comprehensive evaluation mechanism conducted before advancing the design into the subsequent phases of development. The primary focus of the CDR is to ensure that the detailed design aligns with the predefined technical, performance, and operational requirements established during the initial design phase. In our project, which involves the design of avionics and systems for a specialized aircraft, adopting a CDR framework allows for a thorough assessment of key design components. This includes scrutinizing system modelling and control design, trajectory planning, signal processing, and human-machine interface aspects. Each component undergoes rigorous evaluation to ascertain its feasibility, functionality, and performance. The CDR report serves as a consolidated documentation of the project's progress, encapsulating essential elements and providing a snapshot of the conceptual advancement. It aims to achieve several objectives, including capturing the final milestone of the project and evaluating the maturity, completeness, and feasibility of the design. By conducting a comprehensive CDR, the project team can identify any potential issues or shortcomings and make informed decisions to optimize the design before proceeding to the next phase of development.

4. System Engineering Process

4.1 Project Description

Goal: The project goal is to simulate and control the Quasar 3-DOF Helicopter.

4.1.1 Mission Objectives

1. Helicopter should be able to follow the mission profile accurately and sequentially as stated in the mission profile outline.
2. Conduct system modelling and analysis of the 3DOF helicopter.
3. Simulate the system model.
4. Create a trajectory planner.
5. Develop a user interface.
6. Conduct sensor and signal processing.

The overall project is the extension of knowledge from the course AER 715: Avionics and Systems and AER 509: Control Systems from Toronto Metropolitan University.

The following project is being developed by group six students. Starting from Jan 8, 2024, the following project was held using the system engineering method and on a weekly basis meeting the project progress and specific tasks are discussed. By Apr 9, 2024, the project will be finished and demonstrated.

Project Gantt Chart

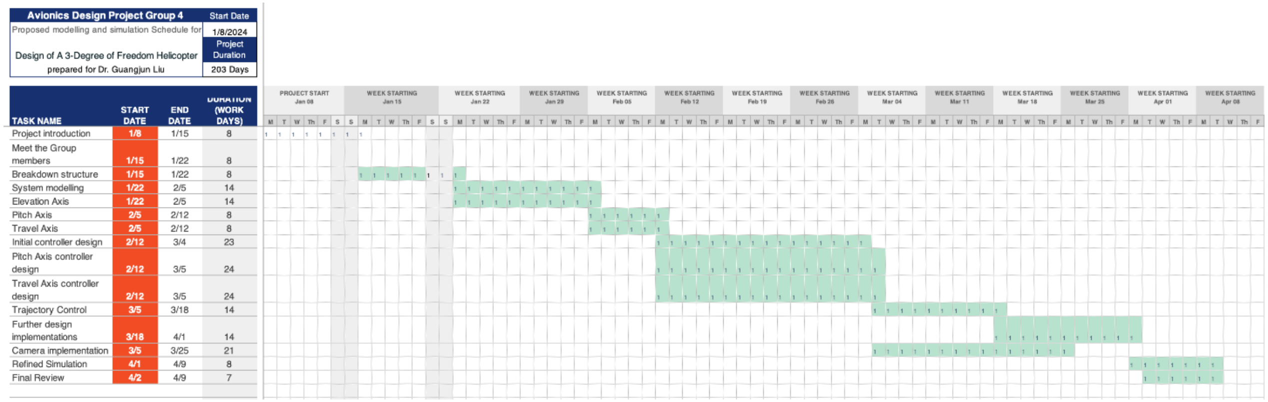

Figure 1: Gantt Chart

Work Breakdown Structure

Table 1: Work Breakdown Str

Topics	Subjects	Student Name	# of Pages
1	Abstract	Robert	1
3	Introduction	Leo	1
4.1	Project Description	Kiran, Simon	1
4.2	Project Gantt Chart	Robert	1
4.3	Work Breakdown Structure	Simon	1
5	Work Package	Everyone	4
6	System Modeling and Control Design	Kiran, Simon	1
6.1	Modelling	Kiran, Simon	1
6.1.1	Elevation Axis Model	Kiran, Simon	3
6.1.2	Pitch Axis Model	Kiran, Simon	2
6.1.3	Travel Axis Model	Kiran, Simon	2
6.2	Controller Design	Kiran, Simon	1
7	Trajectory	Robert, Vijayraj	3
8	Signal processing	Leo	6
9	Simulation Software & Results	All	1
9.1	Controller Implementation	Kiran, Simon	1
9.2	Set Point and Trajectory Evaluation	Robert, Vijayraj	2
10	Human-Machine Interface	Ibrahim	1
10.1	Introduction of HMI	Ibrahim	1
10.2	Design Principle of Visual Information GUI	Ibrahim	2
10.3	Developing GUI in MATLAB-GUIDE	Ibrahim	3

5. WORK PACKAGE (WP)

Project: AER 822 3DOF Design Project					
Work Pack Title:		**HMI-GUI**			**WBS Ref:**
Sheet:	1 of 1				
Scheduled Start: January 15th, 2024				Accountable Manager:	Ibrahim Khan
Scheduled End: Apr 5, 2024		Resources	Toronto Metropolitan University		

Objectives:

· Develop a Graphical User Interface that communicates the machine and that can control the 3DOF helicopter

Inputs:

From Dr. Liu

From Dennis Miklos

From Kiran Paul

From Robert Anadu

From Simon Bae

From Vijayraj Kharod

From Leo Li

Tasks:

- Develop an app using GUIDE
- Allow the design team to use the App to help with simulations
- Gather Ideas for enhancing the experience with GUI

Outputs/Deliverables:

A functioning app that is user-friendly and meets mission requirements.

<table>
<tr><td colspan="5">Project: AER 822 3DOF Design Project</td></tr>
<tr><td colspan="2">Work Pack Title:</td><td>Signal processing</td><td colspan="2">WBS Ref:</td></tr>
<tr><td>Sheet :</td><td>1 of 1</td><td></td><td colspan="2"></td></tr>
<tr><td colspan="2">Scheduled Start:
January 15th, 2024</td><td>Accountable Manager:</td><td colspan="2">Leo Li</td></tr>
<tr><td colspan="2">Scheduled End:
Apr 5, 2024</td><td>Resources</td><td>Toronto Metropolitan University</td><td></td></tr>
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Objectives:

- Develop a clear signal processing system picture for better system control and make the capacity for additional hardware signal input like an IMU or a camera

Inputs:

From Dr. Liu

From Dennis Miklos

From Kiran Paul

From Robert Anadu

From Simon Bae

From Vijayraj Kharod

From Ibrahim Khan

Tasks:

- Develop a block diagram for signal processing.
- Adjust the sensors.
- Create corresponding filters.
- Solve signal-related issues.

Outputs/Deliverables:

Signal filters for fine-tuning.

<table>
<tr><td colspan="5">Project: AER 822 3DOF Design Project</td></tr>
<tr><td colspan="2">Work Pack Title:</td><td>System Modeling and Control Design</td><td colspan="2">WBS Ref:</td></tr>
<tr><td>Sheet:</td><td>1 of 1</td><td></td><td colspan="2"></td></tr>
<tr><td colspan="2">Scheduled Start:
January 15th, 2024</td><td>Accountable Manager:</td><td colspan="2">Kiran Paul, Simon Bae</td></tr>
<tr><td colspan="2">Scheduled End:
Apr 5, 2024</td><td>Resources</td><td>Toronto Metropolitan University</td><td></td></tr>
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Objectives:

· Develop a stable and quick responding controller with the usage of proper control design techniques.

Inputs:

From Dr. Liu

From Dennis Miklos

From Kiran Paul

From Robert Anadu

From Simon Bae

From Vijayraj Kharod

From Ibrahim Khan

Tasks:

- Develop mathematical models that are obtained theoretically and experimentally.
- Develop block diagram to model system.
- Develop a block diagram to progress the control design process.
- Analyze simulation results

Outputs/Deliverables:

Using a control design technique, develop a controller with a reasonable performance.

Project: AER 822 3DOF Design Project					
Work Pack Title:		Trajectory Planning and Control			**WBS Ref:**
Sheet:	1 of 1				
Scheduled Start: January 15th, 2024				Accountable Manager:	Robert Anadu; Vijayraj Kharod
Scheduled End: Apr 5, 2024		Resources	Toronto Metropolitan University		

Objectives:

- Develop a working Trajectory and path control and implement it into the helicopter

Inputs:

From Dr. Liu

From Dennis Miklos

From Kiran Paul

From Simon Bae

From Vijayraj Kharod

From Leo Li

Tasks:

- Helped with the finetuning of the PID controllers.
- Working on trajectory and path planning for the helicopter.
- Working on implementing the cameras with the trajectory control.

Outputs/Deliverables:

Custom trajectory planning Simulink block, that has relevant inputs and outputs

6. SYSTEM MODELING AND CONTROL DESIGN

The section will cover the necessary steps to begin developing a system model and control design of the 3DOF Helicopter. The final model and controller will be simulated to test how well it executes the final flight mission profile.

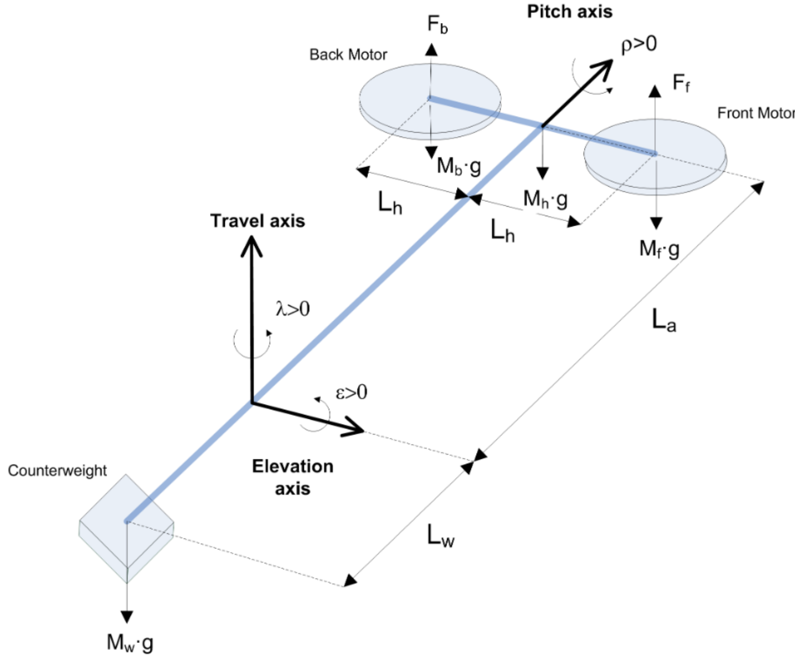

Figure 2: Free-body diagram of the 3DOF Helicopter

The best way to capture the behaviour of the 3DOF helicopter is to consider the rotational dynamics of the system [1]. This way, we can connect how the forces and torques affect the rotational motion about an axis; the axis in question is elevation, pitch, and travel [1]. The following sections aim to describe the non-linear nature of the 3DOF helicopter using mathematical models, which will then be linearized; A linear model must be derived to perform model analysis and apply control design theory [1].

The linear models of the 3DOF Helicopter are derived about the elevation, pitch, and travel axis. To begin, the motion about each axis is considered independent of each other, effectively removing any coupling forces. Therefore, the system can be analyzed as a simple control system with a single input single output also known as the SISO method. In terms of the 3DOF

Helicopter, the input will be the desired angle for elevation, pitch and travel motion and the output being the actual angle of each motion. The motion dynamics for each axis are derived using a differential equation, summing the forces and torques acting on each axis. To simplify and linear the equations various assumptions will be used to achieve an accurate and linearized model for elevation, pitch, and travel.

6.1.1 Elevation Axis Model

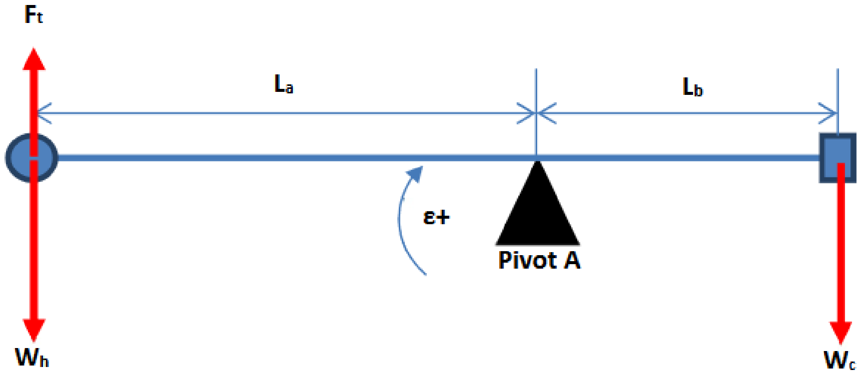

Figure 3: Free-Body Diagram for the Elevation Axis [1]

Based on the free-body diagram in Figure 3, the model for the elevation dynamic can be derived by summing all the torques and inertial forces. The mathematical model of the elevation is described using the differential equation (1). While analyzing the elevation dynamics, this model will consider no pitching such that the elevation torque is only the result of the lift force generated by the front and back motor propellers on the helicopter body. As such, to elevate, the torque generated by motor propellers must be greater than the gravitational torque acting on the propellers.

$$\sum T_i = F_t L_A \cos\varepsilon + W_c L_B \cos\varepsilon - W_h L_A \cos\varepsilon = J_\varepsilon \ddot{\varepsilon} \quad [1] \tag{1}$$

Where,

- L_A and L_B represent the distance from pivot point A to the helicopter body center and the pivot point A to the counterweight center.
- F_t is the total thrust generated by both motor propellers about the elevation axis.
- W_c and W_h is the downward force acting on the helicopter body center and counterweight center.

- J_ε is the total moment of inertia about the elevation axis.

However, as stated above, if the model is analyzed in steady-state flight conditions, based on the small angle theorem:

- As cos approaches 0: $\cos\rho \approx 1$ for the travel model

Effectively linearizing the model in (1). The final model of the elevation axis can be seen down below where T_g is the gravitational torque of the counterweight.

$$J_\varepsilon \ddot{\varepsilon} = F_t L_A - T_g \tag{2}$$

Neglecting gravitational torque, the equation now becomes:

$$J_\varepsilon \ddot{\varepsilon} = F_t L_A \tag{3}$$

Where, F_t can be written in terms of total voltage supplied (V_s) to the helicopter body and the motor-prop force constant (k_f).

$$J_\varepsilon \ddot{\varepsilon} = k_f V_s L_A \tag{5}$$

The final equation for the elevation dynamics is described down below in Equation (6).

$$\ddot{\varepsilon} = \frac{k_f V_s L_A}{J_\varepsilon} \tag{6}$$

By taking the Laplace of (8), the transfer function for the elevation model is:

$$G(s)_{elev} = \frac{\varepsilon(s)}{V_s(s)} = \frac{k_f L_A}{J_\varepsilon s^2} \tag{7}$$

The transfer function above describes the relationship between the voltage supplied to the helicopter and the elevation position as a result.

Where,

- $J_\varepsilon = M_h L_A^2 + M_c L_B^2 = 1.07\ kg - m^2$
- $k_f = 0.14\ N/Volts$
- $L_A = 0.654$

A secondary method was attained to develop an accurate representation of the elevation dynamics. This involved using measured input and output data from the 3DOF Helicopter to construct a mathematical model that approximates the Helicopter's behaviour. This process is known as system identification which was carried out using MATLAB tfest built-in function [2]. The method involves the following:

1. Input Data: An input voltage input of 21.4 V was given to the helicopter. The input voltage was determined based on the voltage required to keep the helicopter at steady-level flight (SLF). Here, the helicopter will take off and begin to oscillate around SLF.
2. Data Collection: Using the Data Recording Block in Simulink, three tests were conducted where the elevation angle and time were collected.
3. Model Estimation: The model was estimated using the test command in MATLAB based on the time-domain data collected. A 2nd and 3rd-degree transfer functions were estimated and then validated to see how accurately the model performed.

Based on this method, the model of the elevation dynamics was found to be the following transfer function:

$$G(s)_{elev1} = \frac{\varepsilon(s)}{V_s(s)} = \frac{0.11}{s^3 + \ 3.9s^2 + 1.4s + 4.6} \tag{9}$$

This transfer function was derived from experimental data and will be used to simulate the elevation. Once the transfer function of the elevation model was found, it was implemented in a block diagram for further analysis and PID tuning as seen below.

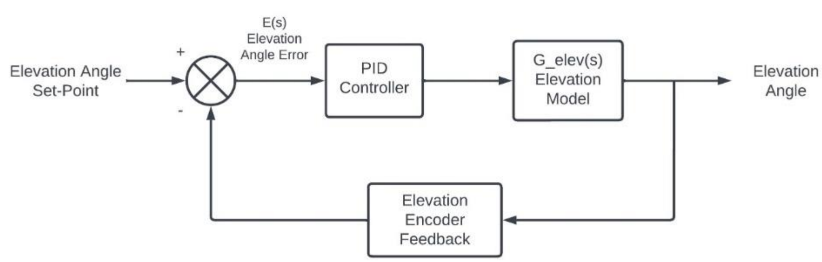

Figure 4: Block Diagram for Elevation Control

6.1.2 Pitch Axis Model

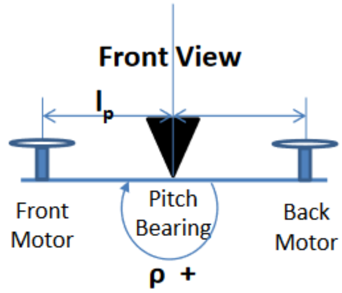

Figure 5: Front View Free-Body Diagram for Pitch Axis [1]

The mathematical model of the pitch dynamics is based on the figure above. Using the same methods seen in elevation, the pitch model was derived. Here, it is evident that a voltage difference in the motors causes the system to pitch. If the lift force generated by the front motor is greater than the back motor, the system will produce a positive pitch, and vice versa. For the pitch model, the system is considered in equilibrium, thus the pitch angle is reduced to zero. The differential equation for pitch is the following:

$$J_\rho \ddot{\rho} = F_f L_P - F_b L_P \tag{10}$$

Where,

- F_f and F_b is the lift force generated by the front and back motor, respectively.
- L_P is the distance between the front and back motor propeller from the pivot point.

The force generated by the front and back motor can be represented in terms of voltage, similar to elevation, where instead a voltage difference results in pitching.

$$J_\rho \ddot{\rho} = k_f L_P (V_f - V_b) = k_f L_P V_d \tag{11}$$

$$\ddot{\rho} = \frac{k_f L_P V_d}{J_\rho s^2} \tag{12}$$

To investigate the stability and performance of the system, the transfer function of the pitch model was found by taking the Laplace of (14). The final transfer function for the pitch model is:

$$G_{pitch}(s) = \frac{\rho(s)}{V_d(s)} = \frac{k_f L_P}{J_\rho s^2} \tag{13}$$

Once the transfer function was found, the block diagram for pitch control was created as an aid for analysis and tunning, as seen in the figure below. Where $G_{pitch}(s)$ is the plant function that represents a transfer function where the pitch angle is the output of the function and voltage is the input to the function. The PD controller is the Proportional-derivative control.

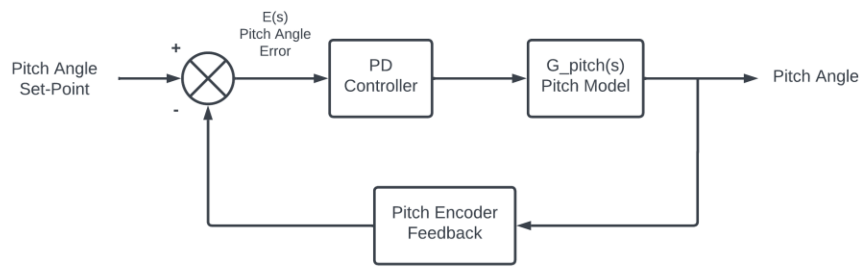

Figure 6: Block Diagram for Pitch Control

6.1.3 Travel Axis Model

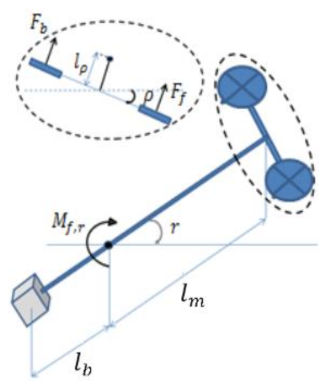

Figure 7: Free-Body Diagram for the Travel Axis [3]

The travel motion occurs when the helicopter body is pitched. The resulting lift force for each motor generates a thrust force resulting in the travel motion of the helicopter body. As such, the dynamic model for travel can be described as the following:

$$J_\lambda \ddot{\lambda} = L_A(F_f + F_b) \sin \rho \tag{14}$$

Since the equation above is a function of voltage and pitch angle, to simplify the expression above, the following approximations are made:

- For small pitch angles, the forces required to keep the helicopter in flight are approximately the total lift force generated by the front and back motor propellers. Thus, as sin approaches 0: $\sin\rho \approx \rho$ [4]
- The elevation angular acceleration is $\ddot{\varepsilon} = 0$. Here, the lift force required to keep the body in the travel motion must be constant [4]. As such, the lift force generated by the motors will equal the force necessary to keep the body in equilibrium or steady-level flight. Since the assumption of elevation angle is 0, it can be said that the voltage required is equal to the operational voltage. Experimentally, the value to keep the helicopter at SLF was 21. 6 V; this can be considered the operational voltage [4].

Therefore, the travel model can be expressed as the following:

$$\ddot{\lambda} = \frac{k_f V_{op} L_A \rho}{M_f L_P^2 + M_b L_P^2 + M_c L_W^2} \tag{15}$$

By taking the Laplace of (21), the transfer function for the travel model is:

$$G_{travel}(s) = \frac{\lambda(s)}{\rho(s)} = \frac{k_f V_{op} L_A}{J_\lambda s^2} \tag{16}$$

Once the transfer function was found, the block diagram for travel control was created as an aid for analysis and tunning, as seen in Controller Design. Where $G_{travel}(s)$ is the plant function that represents a transfer function where the travel angle is the output of the function and voltage is the input to the function. The PD controller is the Proportional-Derivative control.

6.2 CONTROLLER DESIGN

In the previous section, linear models of the elevation, pitch, and travel were developed for 3DOF helicopter. The focus of this section is to design a successful controller that will be implemented via simulation with trajectory planning. Controller gains will be updated in such a way that the deviation between the output of each plant model and the set point is kept at a minimum. The controller gains are estimated using automatic tuning methods in MATLAB and verified via simulation to analyze the performance and tracking of the models.

The controllers developed in the PDR phase failed to produce adequate system responses once each model was combined. The main concern for this section was tuning the pitch and travel system to yield desired system responses, whereas elevation only required further simulation refinement to follow the trajectory plan. For elevation, the controller gains were updated to minimize overshoot and steady-state error.

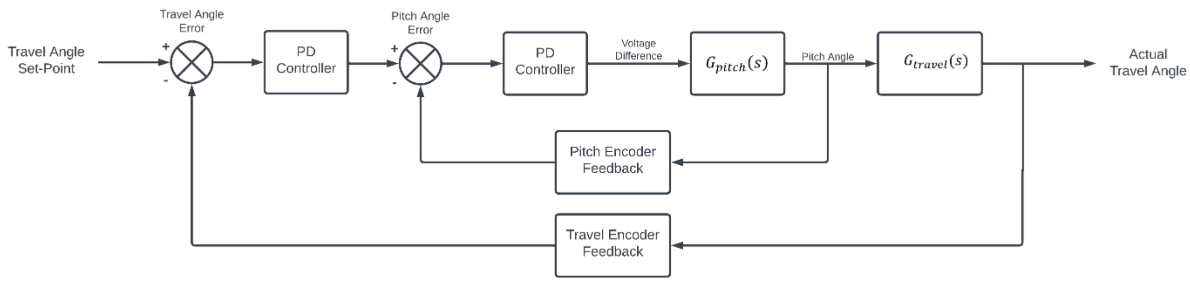

Figure 8: Travel Model Block Diagram

As seen in the figure above, the outer loop generates a travel angle based on the difference between the reference signal and the outer loop output. The travel angle is then used as a reference for the inner loop, which generates the final control signal used to drive the travel model. The helicopter model can only travel if there is a pitch command. As such, it is important to determine the pitch angle to achieve the desired travel angle input as close as possible. The entire dual-loop travel model uses two PD controllers on the measured pitch and travel angle. As a result, new techniques must be used to effectively tune the system, updating the system gains to consider the multiloop system.

A thorough analysis of the controllers will be conducted through simulation to evaluate the trajectory performance before implementing the design in the real-world helicopter. As stated above, the travel and pitch controller was unable to meet performance requirements at the PDR stage. As a result, investigation on potential solutions to solve this instability in the system were investigated via simulation and then tested on the actual helicopter.

The Control System Tuner App within Simulink was used to tune both loops automatically together. Tuning goals were specified for each loop to meet system requirements, which was a settling time of 5 seconds and an overshoot of less than 5 percent. Two tuning goals, looping shaping and reference tracking were set for the inner and outer loop, respectively [5].

The inner loop, pitch control, was tuned and designed based on the “Loop Shaping Goal” found in Simulink. Loop shaping involves designing the loop transfer function to achieve desired performance characteristics, such as stability and robustness [5]. This helps shape the frequency response of the loop to meet specific requirements [5]. Here, bode plots were used to analyze the frequency response of the system to evaluate stability, cross-over frequency, phase, and gain margins. For instance, in the pitch model, a higher bandwidth or cross-over frequency is desirable to achieve a fast rise time; the system will respond quickly to changes in pitch, whereas in the travel plant, stability and accuracy are desired.

The outer loop, travel control, was tuned based on the “Reference Tracking Goal” in the Control System Tuner App within Simulink. Reference tracking refers to the ability of the system to follow a desired angle accurately and promptly; the desired reference signal will be the trajectory path generated based on missions one and two of the project [2]. Setting reference tracking goals such as steady-state error and response time helps determine the required bandwidth for each control loop. For example, in the travel plant, the controller needs to track changes based on the desired travel angle quickly and accurately to maintain stability and achieve desired flight characteristics. The input signal frequency of the system must be within the bandwidth limit of the system to operate without any degradation or attenuation [2].

6.3 Controller Gain Results

The tuned gains for the elevation, pitch and travel PD controller are shown below:

$$\mathrm{K_{p_e}} = 31,\ \mathrm{K_{i_e}} = 2,\ \mathrm{K_{d_e}} = 137$$

$$K_{p_t} = 0.4,\ K_{d_t} = 0.8$$

$$K_{p_p} = 0.5,\ K_{d_p} = 23$$

Where K_p, K_i *and* K_d are the proportional integral and derivative controller gains and the subscript e, p and t denote elevation, pitch, and travel models.

7. TRAJECTORY PLANNING

Path and trajectory planning is extremely useful in the aerospace industry. It defines the systematic movement of a robot from one location to another in a controlled manner. Although a path shows the sequence of robot configurations in a specific order, it does so irrespective of the time it takes while a trajectory focuses on the time it takes each part of the path to finish. Effective path and trajectory planning can be achieved by the utilization of both kinematics and dynamics thus ensuring precise and efficient movement execution in various environments and scenarios.

Given the current position of the helicopter in degrees and a target waypoint, a path needs to be computed that smoothly accelerates and decelerates to the target. It can be further given an initial and final velocity and acceleration. Using the path planner as a high-level controller and the PID controller as a lower-level path follower allows for much larger and multipoint trajectories without having to finely tune the PID directly for large steps.

The trajectory planner is a quintic polynomial trajectory solver that takes the following inputs,

q_0 - Start position
q_f - End position
v_0 - Start velocity
v_f - End velocity
a_0 - Start acceleration
a_f - End acceleration
t_0 - Start time
v_a - Average velocity

The average velocity is used to calculate the end time. This allows the user to define an easily understandable parameter without having to guess the end times.

$$t_f = \frac{|q_0 - q_f|}{v_a} + t_0 \tag{17}$$

The coefficients for the quintic polynomial,

$$\theta(t) = x_0 + x_1 t + x_2 t^2 + x_3 t^3 + x_4 t^4 + x_5 t^5 \tag{18.1}$$

Given the initial conditions, the coefficients can be found by solving the following system of equations,

$$\begin{bmatrix} q_0 \\ v_0 \\ a_0 \\ q_f \\ v_f \\ a_f \end{bmatrix} = \begin{bmatrix} 1 & t_0 & t_0^2 & t_0^3 & t_0^4 & t_0^5 \\ 0 & 1 & 2t_0 & 3t_0^2 & 4t_0^3 & 5t_0^4 \\ 0 & 0 & 2 & 6t_0 & 12t_0^2 & 20t_0^3 \\ 1 & t_f & t_f^2 & t_f^3 & t_f^4 & t_f^5 \\ 0 & 1 & 2t_f & 3t_f^2 & 4t_f^3 & 5t_f^4 \\ 0 & 0 & 2 & 6t_f & 12t_f^2 & 20t_f^3 \end{bmatrix} \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} \tag{18.2}$$

The output position can then be sampled at any point if it is within the start and stop time boundary. At any point outside, the program will enter a hold mode by using a clamp function between t_0 and t_f ; unless a specific mode, such as landing, is specified. Two separate trajectories are calculated independently for the travel and elevation axis using the same method.

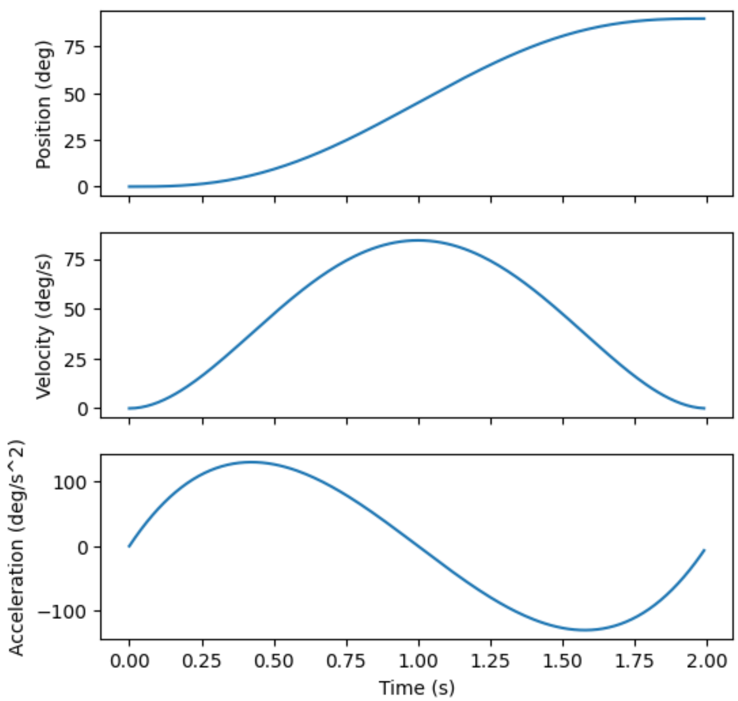

Figure 9: 5th Order Trajectory From 0 To 90 deg

If the controller is unable to follow the path with enough accuracy, the velocity can be reduced to give the controller more time and less acceleration to correctly follow it.

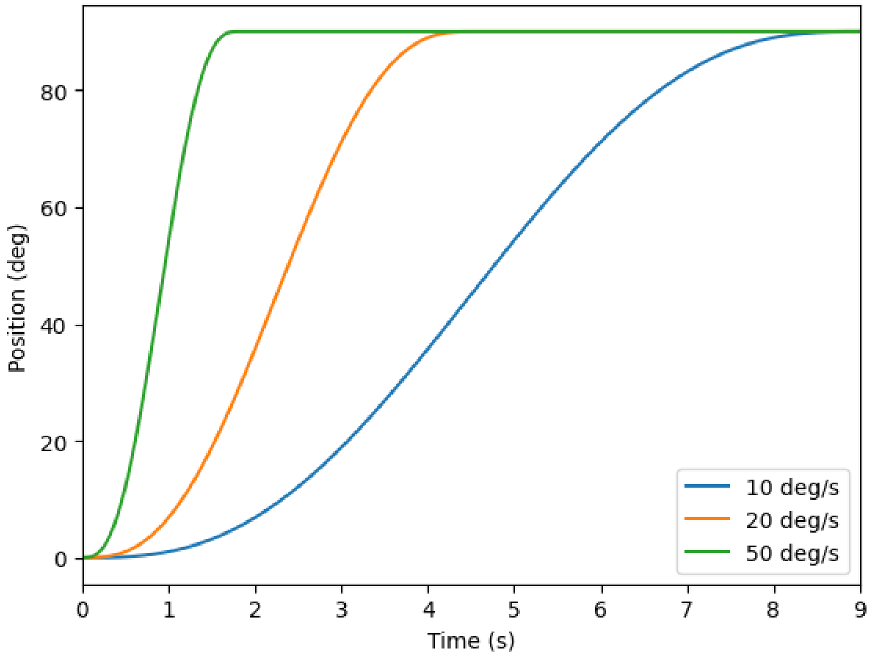

Figure 10: Effect of velocity on trajectory

The multipoint trajectory is calculated by looping over all waypoints in sequence and calculating each segment separately.

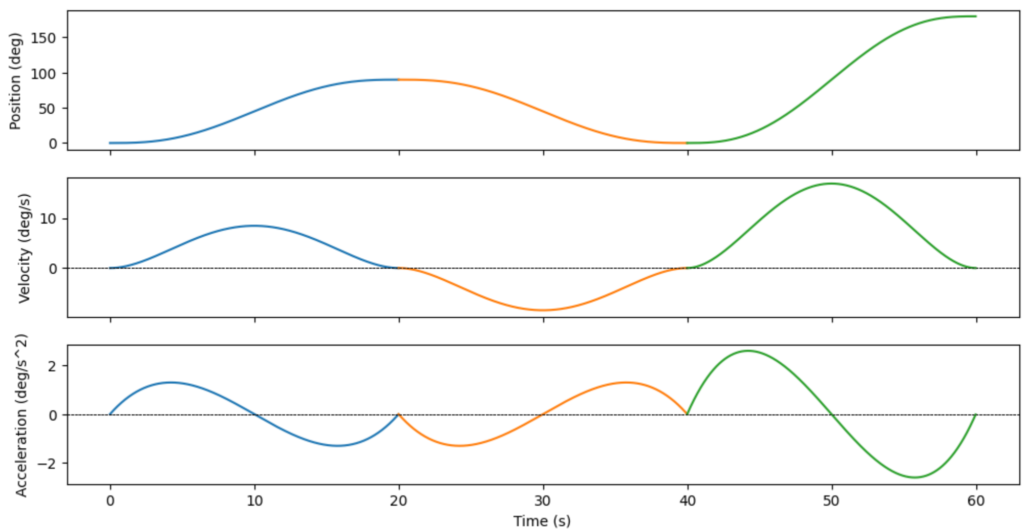

Figure 11: Multipoint 5th-Order Trajectory

The start and end velocity/acceleration are assumed to be 0, so the helicopter comes to a complete stop at every waypoint, but it could be set to an average velocity to quickly transition from waypoint to waypoint, however, this is not required for the mission. Finally, instead of explicitly providing a start and end time, a preset average velocity could be used to calculate the times to provide more consistency across different lengths of runs.

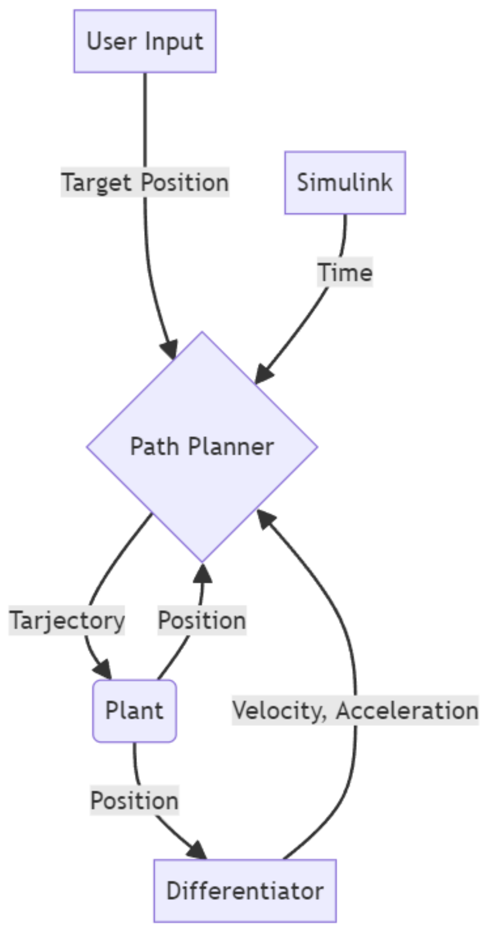

Figure 12: High-level trajectory planner overview

7.1 MISSION 1 PATH PLANNING

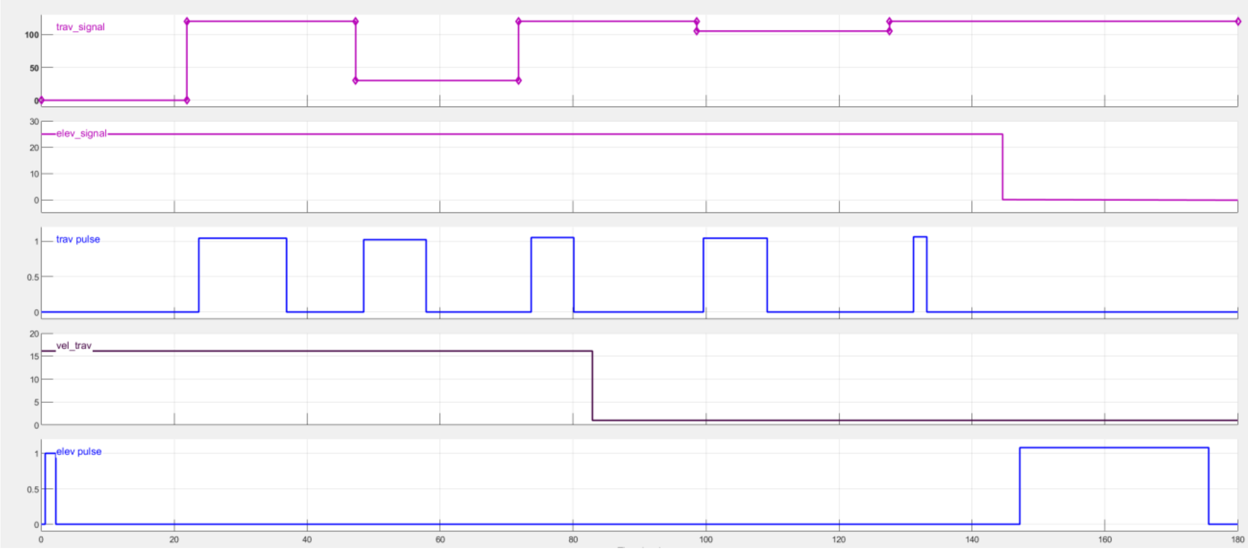

Figure 13: mission 1 signal builder waypoint delivery

Mission 1 uses the signal builder as a tool to provide waypoints and update pulses for the trajectory planner.

Table 2: Mission 1 signal description

Signal	Description
trav_signal	Target position for travel
elev_signal	Target position for elevation
trav_pulse	Pulse to update travel position in the trajectory planner
vel_travel	Target travel velocity
elev_pulse	Pulse to update travel position in the trajectory planner

By using this method, each component and velocity of the trajectory can be controlled and updated at specific points. The trajectory planner is capable of changing trajectories at any point and does not require the completion of the previous trajectory while fully compensating for the current velocity and acceleration of the helicopter.

7.2 Mission 2 Path Planning

Mission 2 uses a mixture of quintic trajectories from the trajectory generator and simple linear trajectories with a constant velocity when the endpoint is not known, specifically when the helicopter is finding the board with the IR sensor and when the helicopter is drilling.

The waypoint generator for the second mission is based on the state of the mission the helicopter is in and the inputs from the users and sensors on the drill.

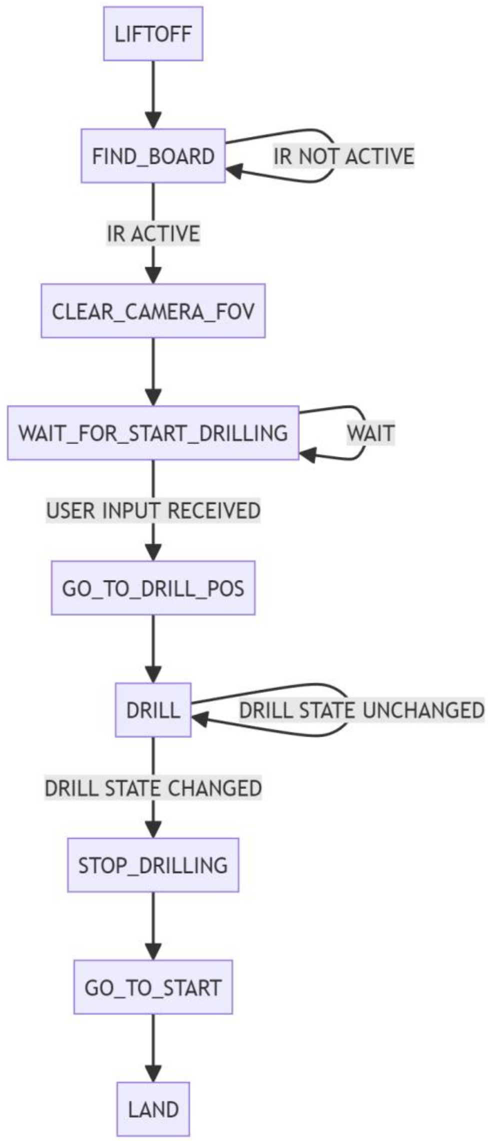

Figure 14: Mission 2 state and path planning

8. SIGNAL PROCESSING

Signal processing plays a vital role in the 3-degree-of-freedom (3DOF) helicopter control system, facilitating communication between hardware components and the computer. Sensors capture physical information, which is then transformed into digital data through signal processing. This digitized information serves as feedback for the computer, enabling the formation of a closed-loop system. Based on this feedback, the control system adjusts parameters to effectively manage the helicopter's motion and stability. Thus, signal processing acts as a bridge between the physical world and the digital control system, facilitating real-time adjustments for optimal performance.

8.1 SENSOR SYSTEM

Here is the overall configuration of the sensor system:

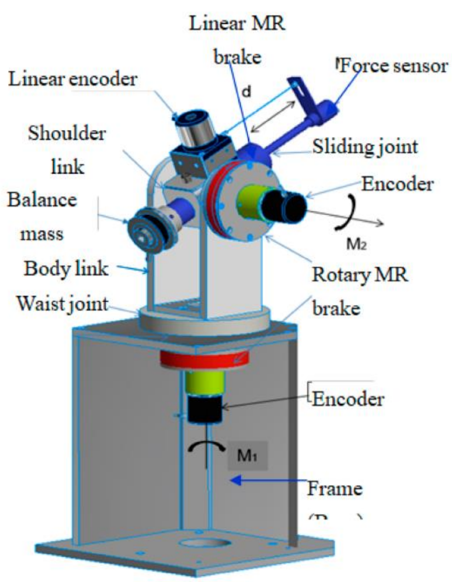

Figure 15: Sensor Configuration of 3DOF Helicopter

This figure shows the overall configuration of how the sensors are mounted on the device. They consist of 2 rotary encoders, 1 linear encoder, and 1 force sensor.

Encoders serve as the primary sensors in our system, providing essential positional information necessary for data processing. Mounted on the shafts controlling elevation, pitch, and travel direction, three rotary encoders utilize light or other mediums to detect shaft position.

Here is how the encoders work: Operating on the principle of light reflection, these encoders feature a disk attached to the shaft with transparent segments. When a light source is emitted onto these segments, the reflective light is detected by a photodetector, akin to sensors using infrared technology on moving objects. Thanks to the rapid speed of light, any error introduced is negligible. Consequently, the output, comprising a series of position data, facilitates calculations of shaft speed and enables precise control of the system.

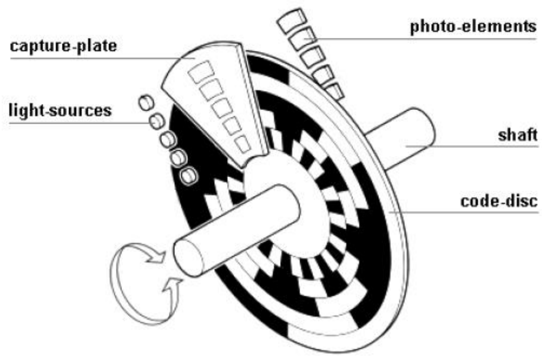

Figure 16: Light Encoder Demonstration

The figure above shows a demonstration of a light encoder with a description of the detailed components.

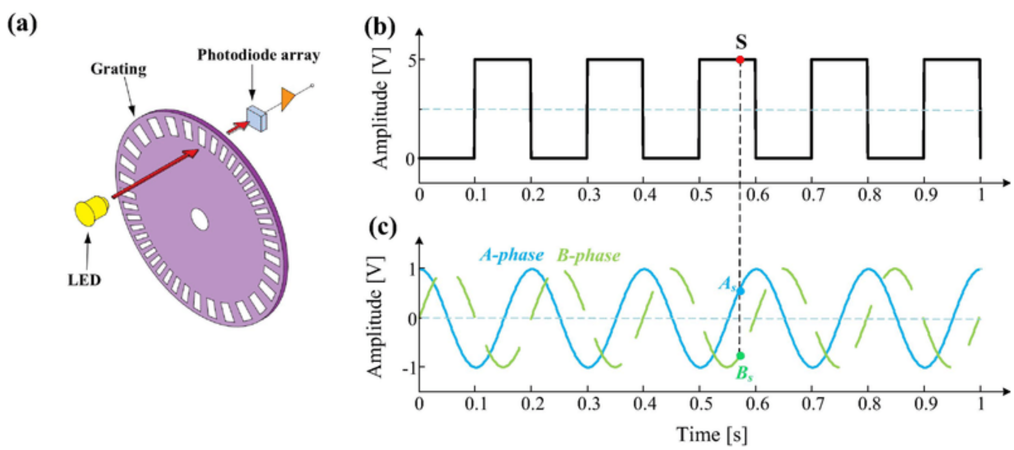

Figure 17: Signal Output of an Optical Encoder

The signal output of optical encoders typically exhibits two main forms: periodic waves and square waves, as illustrated in the accompanying diagrams. The periodic wave features a period corresponding to the rotation or movement of the encoder shaft, while the square wave comprises alternating high and low voltage levels. The rising and falling edges of the square wave correspond to transitions between light and dark regions on the encoder disk, making it suitable for increment encoders. In the helicopter system, where precise position and velocity data are crucial, the encoder signal provides direct information about the absolute position of the shaft. This signal may consist of multiple digital bits or an analog signal representing the angular position.

There is a secondary system to support the encoder system. The Active Disturbance Rejection Control (ADRC) encoder is mounted between 2 motors. This sensor is utilized within a control system that implements an active disturbance rejection control strategy. In our system, the encoder provides feedback on the position or motion of a controlled object. The ADRC controller then utilizes this information, along with disturbance estimates from the ESO, to effectively reject disturbances and maintain control performance.

8.2 SIGNAL INTEGRATION

Signal integration in helicopter control is the process of combining and processing various sensor inputs and control commands to ensure stable and precise flight control. In a real helicopter control system, a wide array of sensors is utilized, including gyroscopes, accelerometers, airspeed indicators, and more. These sensors provide critical data on the helicopter's orientation, velocity, altitude, and other parameters.

In our 3-degree-of-freedom (3DOF) helicopter system, a specific algorithm is implemented to integrate the signal outputs of the encoder systems. These encoders measure the position and motion of the helicopter's elevation, pitch, and travel axes. By processing the data from these encoders, along with inputs from other sensors and control commands, the algorithm calculates the appropriate control signals to adjust the helicopter's position and orientation as desired.

The signal integration process involves combining sensor data, filtering out noise and disturbances, and using sophisticated algorithms to generate precise control signals. This ensures that the helicopter responds accurately to pilot inputs and maintains stable flight even in challenging conditions. To track the orientation of our 3DOF helicopter, this differential equation is used:

$$\dot{R}(t) = R(t)\Omega(t) \tag{19}$$

While R(t) is the current orientation of the craft (represented as a rotation matrix defined by the rotations through the Roll/Pitch/Yaw Euler angles)

Rearrange and solve:

$$R(t) = R_o(t) \cdot expm(\int_0^t \Omega(t)dt) \tag{20}$$

R0(t) is the known initial orientation of the chopper. Omega(t) is the skew-symmetric form of the angular rate vector (take the body rates measured by the rate gyro and form this matrix):

$$\Omega(t) = \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix}$$

Result:

$$R(t + \Delta t) = R_o(t) \cdot expm(\int_t^{t+\Delta t} \Omega(t)dt) \tag{21}$$

Next, a complementary filter was implemented to enhance parameter estimation by integrating data from two separate sensors, each excelling at different frequency ranges, to provide a more accurate measurement. Typically, one sensor, such as an accelerometer, offers reliable estimates at low frequencies, while the other, like a rate gyro, is proficient at higher frequencies but may have biases. In the context of a 3DOF helicopter, the utilized filter type is the Passive Complementary Filter with Bias Correction, designed specifically to rectify biases inherent in rate gyros. This correction is achieved by leveraging additional sensors like accelerometers and magnetometers to estimate the helicopter's state, ensuring improved accuracy in measurements.

This equation is the result after the filter was applied:

$$R(t + \Delta t) = R_o(t) \cdot expm(\int_t^{t+\Delta t} (\Omega(\mathrm{t}) - \hat{b} + K_p\omega)dt) \tag{22}$$

While omega is formed via an anti-symmetric projection operator ‘*’ and is an estimate of the rotational position error. The Kp scales omega.

Lastly, to determine the bias of the gyro rate, this equation is applied:

$$\hat{b} = -\int_{t}^{t+\Delta t} (K_i \omega) dt \tag{23}$$

Finally, the control block diagram implemented in the system based on the integration above is shown below:

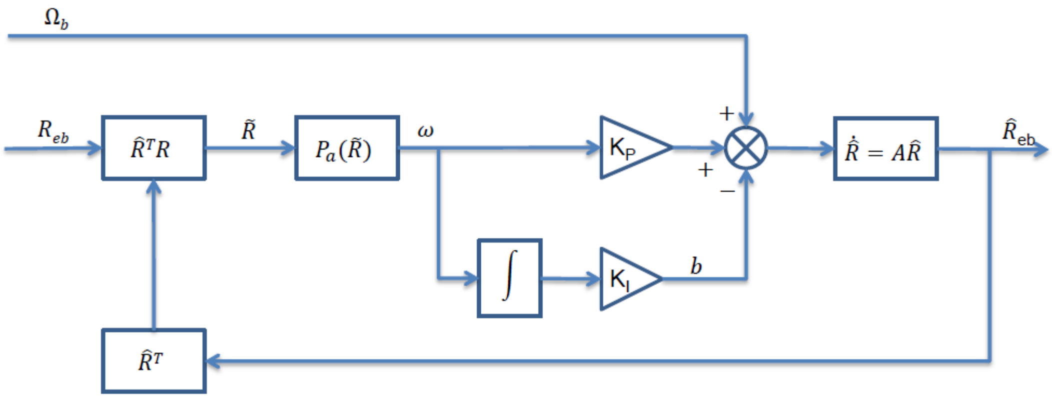

Figure 18: Nonlinear Signal Filter Integration

This matrix below shows the calculation to get the global or navigation rates.

$$\begin{bmatrix} \dot{\varepsilon} \\ \dot{\rho} \\ \dot{\tau} \end{bmatrix} = \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} 1 & sin\phi tan\theta & cos\phi tan\theta \\ 0 & cos\phi & -sin\phi \\ 0 & sin\phi / cos\theta & cos\phi / cos\theta \end{bmatrix} \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix}$$

Lastly, the implementation in Simulink is shown below:

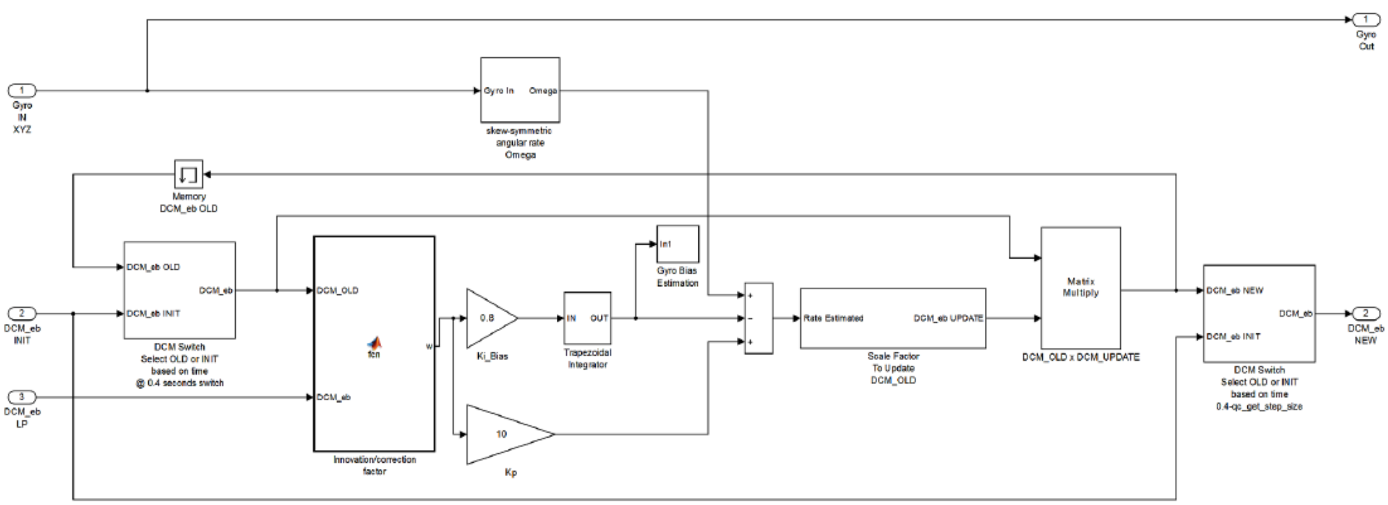

Figure 19: Signal Integration in Simulink Implementation

To sum up, signal integration in helicopter control, particularly through the application of Complementary filters, stands as a cornerstone in enhancing flight stability and precision. By amalgamating data from two distinct sensors, each excelling in different frequency ranges, such as accelerometers for low frequencies and rate gyros for higher frequencies albeit prone to biases, a more comprehensive and accurate estimation of parameters crucial for flight control is achieved. This harmonious blending of sensor outputs not only ensures smoother flight operations but also mitigates the impact of inherent sensor limitations, thereby significantly bolstering the safety and efficiency of helicopter navigation in various operating conditions.

8.3 SIGNAL PROCESSING

The purpose of signal processing is to aid the control system in noise reduction and feature extraction. By reducing unwanted noise from the environment, system performance and signal output quality can be improved. Additionally, feature analysis can help recognize patterns in the system output, enabling fine-tuned adjustments to be made.

The objective of signal processing in the 3 DOF helicopter system is to transmit the pitch, elevation, and travel data from encoders to the computer and generate feedback to the helicopter and form a control system.

To measure the signal, encoders mounted on the helicopter capture the respective movements to get the data of positions and velocities. These encoders convert mechanical motion into electrical signals. Then, the electrical signals from the encoders undergo signal conditioning processes to improve their quality and compatibility with the computer system. This can include amplification, and filtering to remove noise. The conditioned signals are then transmitted to the computer using suitable communication protocols. This transmission is through cables. Specifically, the communication between the 3 encoders and the computer is mainly through wired connection. these 3 cables are for the sensor system:

Figure 20: Sensor-System Communication Cables

The first one carries the encoder signals between an encoder connector and the data acquisition board with four sub-branches: power supply, ground connection, and 2 channels. The second one carries analog signals (joystick, plant sensor) to the UPM. The last one serves like the second one but is used on the terminal board.

Upon reaching the computer, the transmitted signal is received by the computer interface modules. These interface modules are designed to interpret the incoming signals and convert them into digital data that can be processed by the computer. The digital data representing pitch, elevation, and travel information are then processed by *Simulink*. This processing involves tasks such as real-time monitoring of the helicopter's orientation and position, implementing control algorithms to adjust flight parameters based on pilot inputs or autonomous control systems, and logging data for analysis and feedback purposes. Based on the processed data, control commands may be generated to adjust various aspects of helicopter operation. These commands are then transmitted to the appropriate actuators (e.g., servo motors or hydraulic systems) to execute the desired actions, thus closing the control loop. And here is the actual plant in *Simulink:*

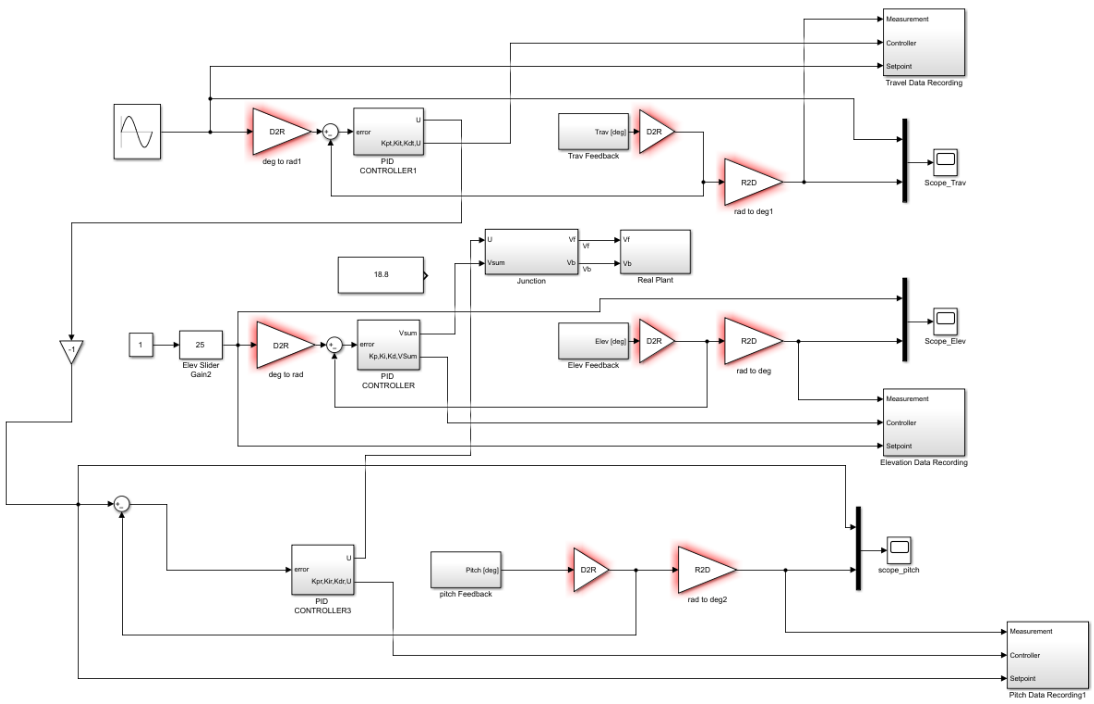

Figure 21: Real Signal Processing Configuration

The schematic above illustrates the data processing flow. Starting with fundamental block logics, the red triangular blocks convert between angles and degrees, the data recording blocks record relevant data, scopes visualize real-time control variables, and plus and minus circle blocks construct the closed feedback loop. PID controller blocks contain the controlling functions, and constant and wave blocks serve as inputs.

8.4 SIGNAL ANALYSIS AND FILTER DESIGN

The transfer function to be optimized is shown below：

$$G(s)elev1=\varepsilon(s)Vs(s)= 0.11s3+ 3.9s2+1.4s+4.6$$

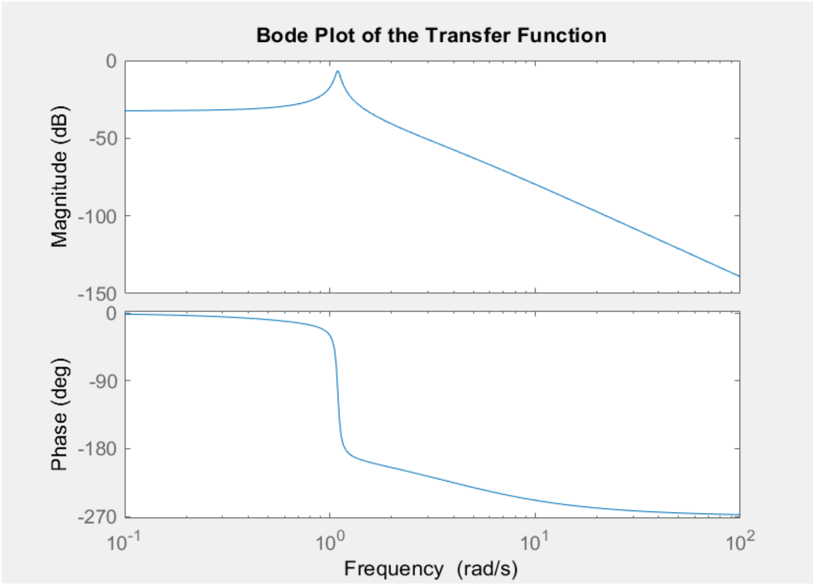

Figure 22: Bode Plot of the Transfer Function in Frequency Domain

The figure above aids in analyzing the behaviour of a transfer function by providing insights into how the system responds to different frequencies. The top plot, a magnitude plot, displays the magnitude of the transfer function (in decibels) as a function of frequency. This plot helps in understanding how the system amplifies or attenuates input signals at different frequencies. Peaks or troughs in the magnitude plot indicate resonant frequencies or points of instability, respectively. The bottom plot, a phase plot, illustrates the phase shift (in degrees) of the transfer function as a function of frequency. It helps in understanding the time delay or advance introduced by the system for different frequencies. Phase margins and phase crossover frequencies can be determined from the phase plot, providing insights into system stability and transient response.

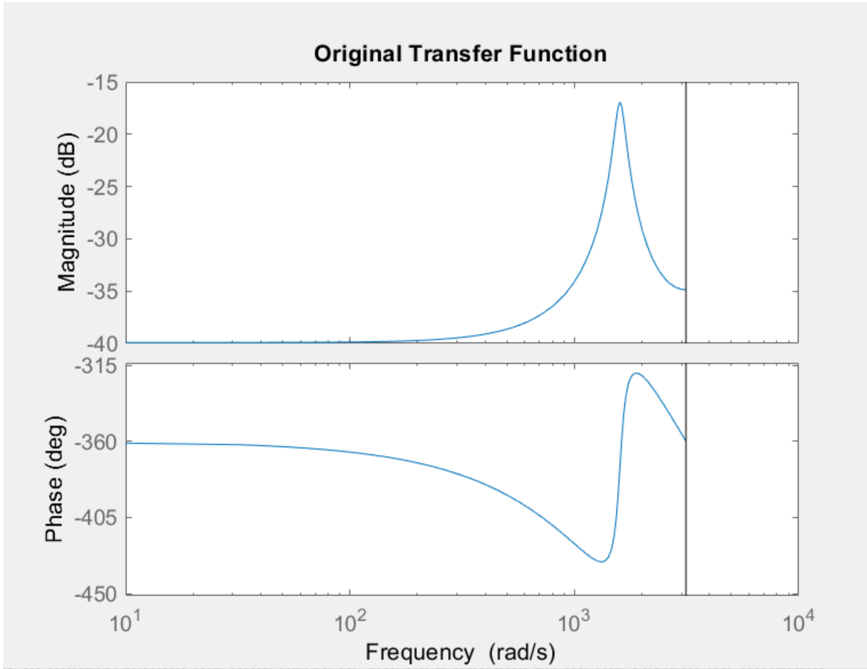

Figure 23: Rescaled Graph of the Transfer Function

To visualize the characteristics of the transfer function, the first thing to do is to rescale the graph by rescaling the axes. The figure above is the rescaled figure of the previous one. The effective domain was extended to 5000 and the range of the magnitude was adjusted from -40 to -15. Due to the shape of the curve, logarithmic scaling was not necessary.

The most signal-related item in the main control plant, a *Simulink* schematic, is the pulse generator block. The Pulse Generator block generates square wave pulses at regular intervals. The block waveform parameters, amplitude, pulse width, period, and phase delay, determine the shape of the output waveform. The following diagram shows how each parameter affects the waveform [4]. The pulse generator block helps visualize the signal with respect to time.

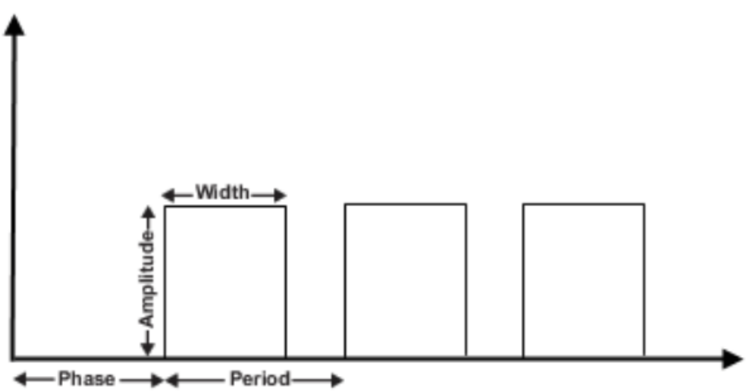

Figure 24: Pulse Generator Block

The figure above illustrates how a pulse generator block process signal, to be specified with the relationship of the pulse width and the period, follows either one of the two situations of the following equations:

$$Period \cdot \frac{PulseWidth}{100} = 100 \tag{24}$$

$$Period \cdot \frac{PulseWidth}{100} = Period \tag{24}$$

The Signal Builder block in *Simulink* serves as a tool for creating and editing complex waveforms or time-based signals without the need for writing *MATLAB* code. Users can define signal waveforms graphically using this block. Below is an example of the modifiable signal input in the Signal Builder block:

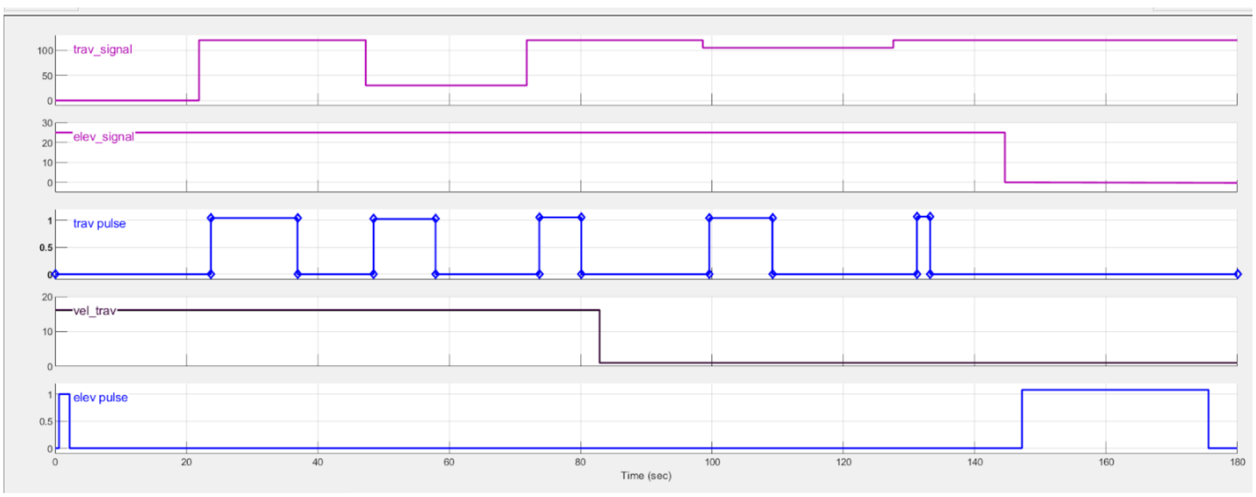

Figure 25: Signal Builder in Simulink

The Signal Builder block in *Simulink* functions similarly to other waveform tools in engineering software, enabling users to modify input signals for parameters like elevation and travel over time. Users can adjust the signal magnitude within different ranges. For instance, in the provided figure, the travel signal has a range of 100, whereas the elevation signal has a range of 30. The magnitude of the signal serves as an instruction guide, specifying the system's actions at specific times.

8.4 Performance and Result

In the working environment of the helicopter system, noise poses a significant challenge to the sensors, primarily due to the vibrations from the helicopter itself and the drill. To mitigate this issue, a notch filter is employed. Notch filters are commonly used to eliminate unwanted interference or noise from a signal, particularly humming or buzzing noises caused by electrical interference, ground loops, or other sources. In this case, the double vibration caused by the drill and the two motors is the targeted interference.

The notch filter applied in this system was designed using *MATLAB*. The design procedure involves specifying the transfer function to be applied to the filter, along with the frequency and sampling time. A plot for the Notch filter was created and the Bode plot after applying the filter as shown below.

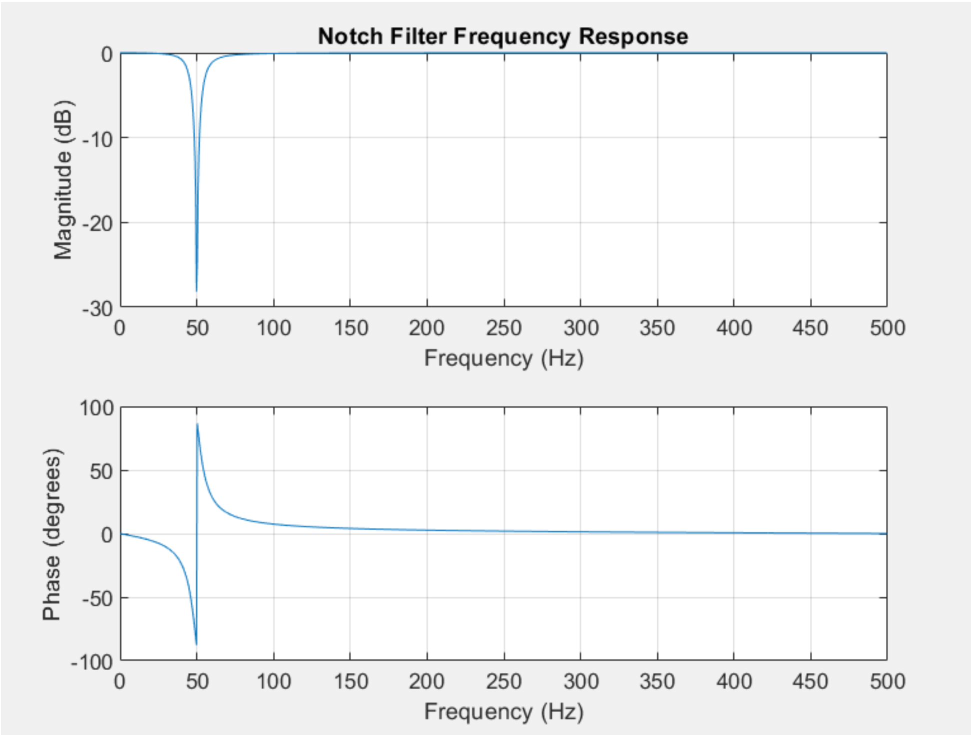

Figure 26: Notch Filter Visualization

And this figure shows the final design of the Notch filter. Like other Bode plots, the notch frequency is set to 50 Hz, indicating the frequency that needs to be attenuated. The bandwidth is set to 10 Hz, defining the range around the notch frequency where the attenuation will take place. The normalized notch frequency and bandwidth relative to the Nyquist frequency are 25 Hz. The reason for setting the notch frequency to 50 Hz is based on the frequency of the drill noise. Since the system already has its own filter for noise from the double motors, the focus of the filter design is the additional noise from the drill. A plot for the Notch filter and the Bode plot was created after applying the filter as shown below:

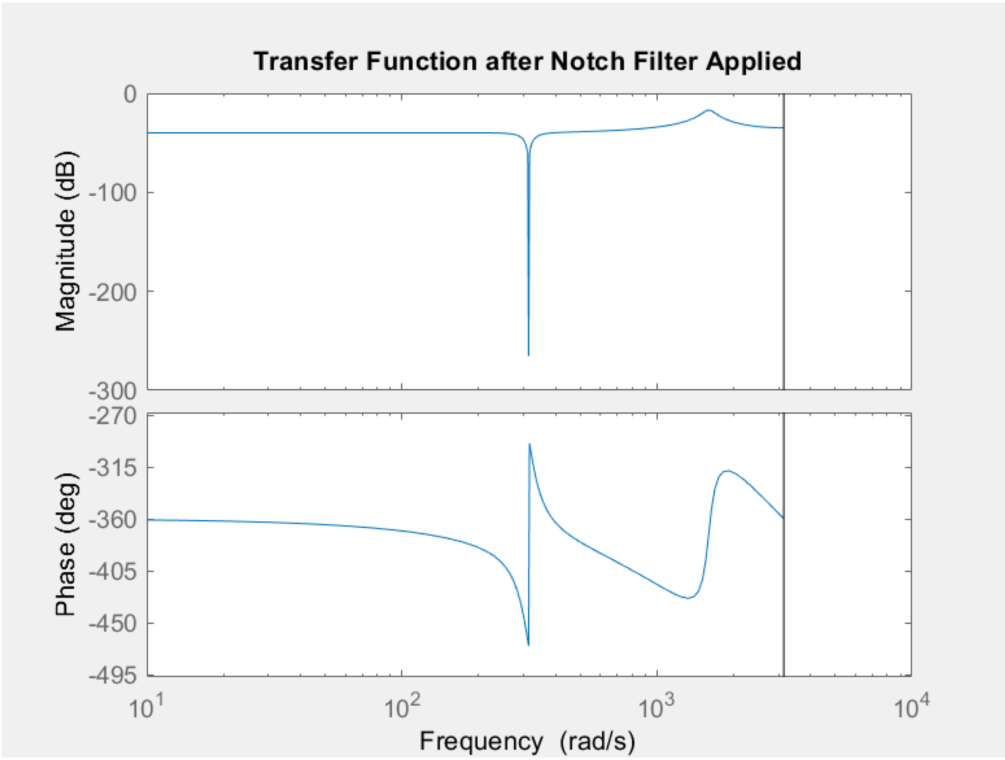

Figure 27: Transfer Function Performance After the Filter is Applied

After applying the Notch filter, the 50 Hz noise was attenuated, allowing the system to mitigate the interference caused by the drill.

In summary, signal processing plays a crucial role in process visualization, sensor-computer optimization, and trajectory control. Converting processing components such as transfer functions into Bode plots facilitates visualization and analysis of system features, enabling optimization through fine-tuning. Fine-tuning in trajectory control aims to generate smoother motion, and reduce overshoot, settling time, and steady-state error.

8.5 Camera Implementation

The main goal concerning the camera function was to determine the relative location of the red blob based on the black blob and calibrate this location to degrees for real-world application with the helicopter. This location was then used to drill at the desired site in the second half of mission two. The camera live feed required signal processing to ensure the Simulink model of the helicopter understands the incoming information being sent from the camera client and the drill.

Signal processing involves signal capture, processing, analysis, and interpretation. A signal is referred to as a physical variable that varies with respect to time or space [6]. Several strategies and procedures were used in signal processing for the camera implementation in the final design to identify and extract useful information. For the context of this report, the focus was on digital image processing which uses a digital computer to process digital images sensed by the camera. Here, digital images are made up of elements with spatial coordinates x and y, and amplitudes also known as intensity or grey level at finite, discrete qualities [7]. These elements are called pixels. The main goal was to extract useful information from the video data using the techniques that will be mentioned in the next section [7].

8.6 BLOB ANALYSIS

Blob analysis is a technique used in image processing to identify and analyze areas of interest (which in this case are the three black blobs and one red blob) from the video feed.

8.6.1 Image Processing

Colour segmentation was used to create a clear distinction between the black and red blobs. It was crucial to identify and collect data for the red blob for the drilling mission while using the fixed black blobs as a reference. A filtering technique was applied to the incoming video feed to filter the blobs based on colour and luminance [7]. The video feed was separated into its primary colours, RGB and then converted to the YCbCr colour space. This gave a clear distinction between the red and black blobs since the Cr channel and the R channel will exhibit distinct variations as seen in Figure 28 [7]. The Y channel displays the luminous component of the video feed as seen in the far-right of the figure below representing the brightness of the image based on a grayscale [7]. Markers were also added to the video feed as visual feedback from the blob analysis.

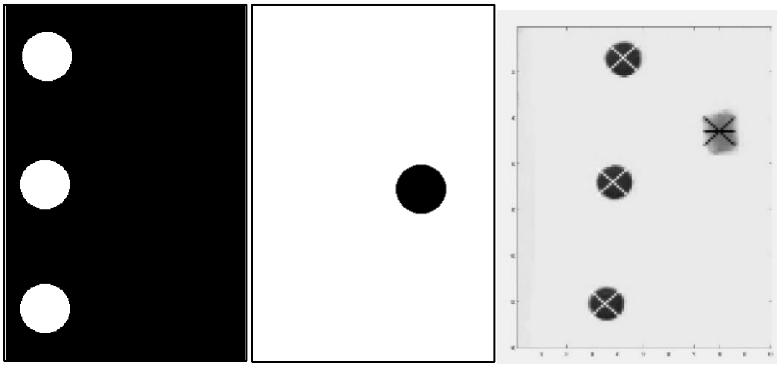

Figure 28: Colour Segmentation results.

After colour segmentation, the binary image is fed into the blob analysis block which connects groups of pixels with similar characteristics, such as intensity, colour, or shape known as thresholding seen in the figure below of the Simulink camera client model [7]. The binary image from the Cr and R channels was then multiplied by a threshold value of 255 to scale the binary image to ensure a sharp contrast for easier visual interpretation as seen in the images above [7]. The final Simulink model for the camera client can be seen in down below.

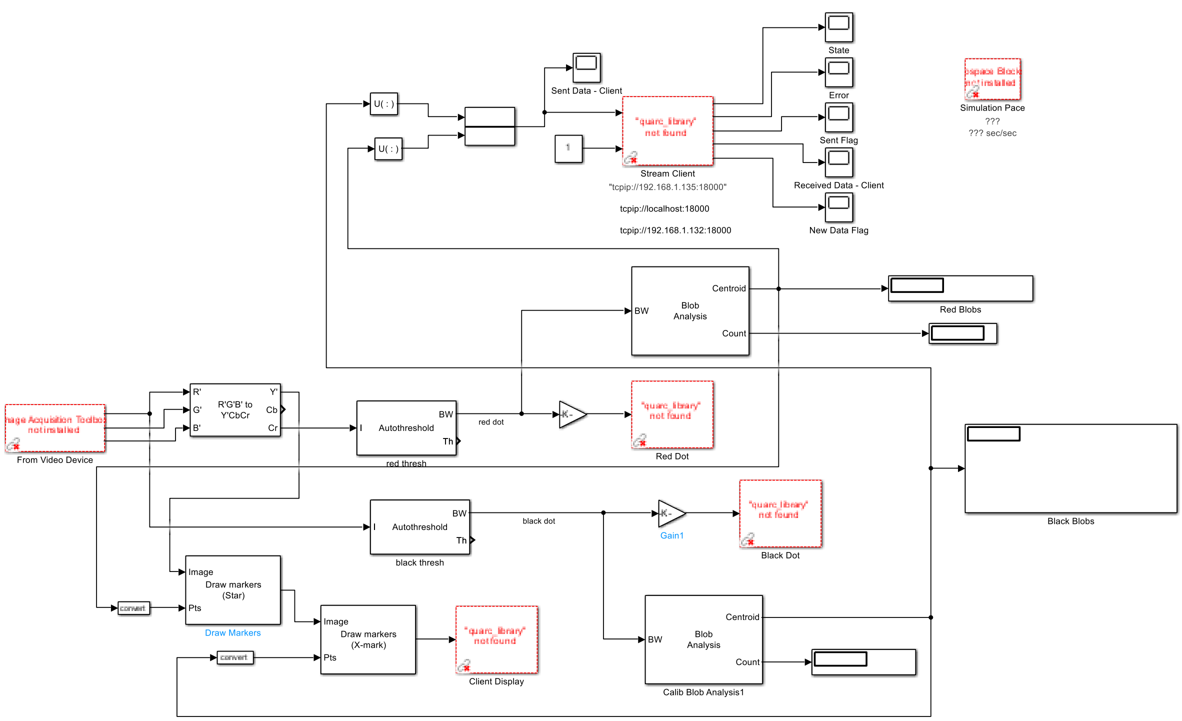

Figure 29: Blob Analysis

8.6.2 Calibration: Pixel to Degrees Conversion

Each blob is then identified in the binary image where the "Blob Analysis" Simulink block in Figure 29 computes the centroid of each blob. Since the mission is focused on identifying the red blob for analysis and drilling, it was important to create a method to filter the blobs on the whiteboard. The centroid of each blob is not enough to establish coordinates for the helicopter to move and drill at the red dot site. As such, calibration of the pixel centroids must be determined to create a relationship between pixel coordinates to degree commands.

Calibration involved determining the mapping between pixel coordinates in the camera image and the real-world coordinates which correspond to the elevation angle of the helicopter. The real-world coordinates of the black blob centroids were calculated in degrees based on the helicopter setup. The helicopter was manually moved to each centroid location where the angles were noted. It was found that each blob was separated by 10 degrees with the first blob being located at 30 degrees. This allowed us to create a relationship between pixels and degrees which was found to be:

$$Degrees\ per\ pixel = \left(\frac{20}{B_{3_y} - B_{1_y}}\right)$$

Where B_{3_y} *and* B_{1_y} are the y-axis pixel location of the 3rd blob (lowest) and the 1st blob (highest) and 20 degrees represent the difference in degrees of the highest blob to the lowest blob.

By taking the difference between the red blob and the first blob's y-axis location in pixels and multiplying it by the degree per pixel relation, the angle of the red blob can be determined as seen below.

$$Red\ dot\ Angle = \left(\frac{20}{B_{3_y} - B_{1_y}}\right) \times (R_y - B_{1_y}) + 13.75$$

Note: An offset of 13.75 degrees had to be applied to the equation to account for the difference from SLF to the first black blob.

As such, the red dot location was determined based on the relative location of the black blobs. Once the blob centroid data was sent to the server, a custom function was created to convert the y- y-coordinate of the pixel data to degrees in real time as seen in the Simulink model found in Figure 30.

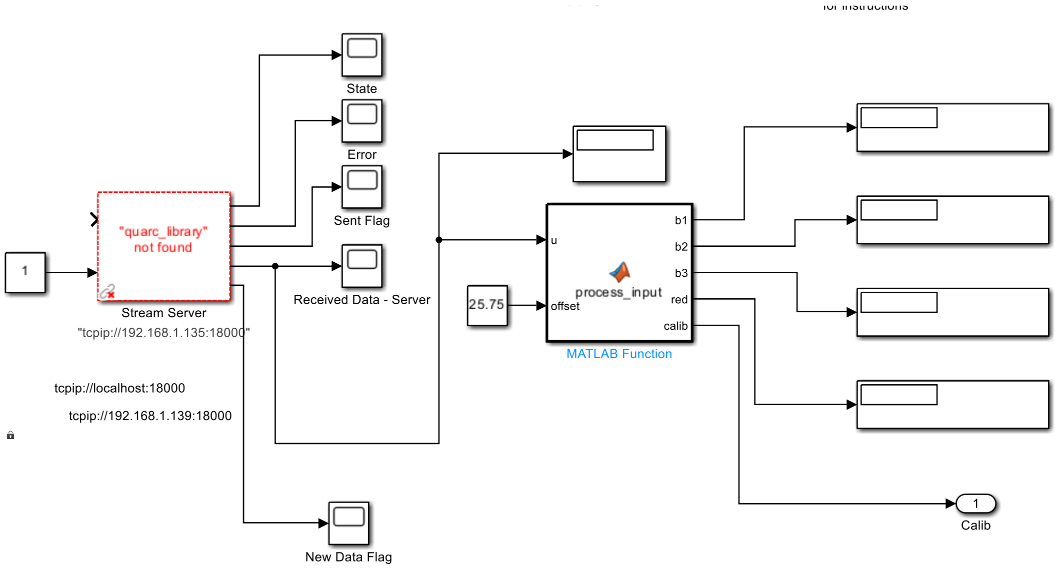

Figure 30: Simulink Model for Calibration

The signal processing tool used in this section aided in mission two by identifying the red blob and sending the coordinates of the blob to the trajectory planner. These coordinates are used as waypoints to generate a path for the helicopter to fly to the drill site.

9. SIMULATION SOFTWARE & RESULTS

9.1 SIMULATION RESULTS WITH TRAJECTORY CONTROL

Once the controller parameters were tuned to optimize performance, validation of the controller was conducted via simulation before testing on the actual helicopter. The final simulation model of the helicopter with trajectory planning developed in Simulink can be seen in below. The results of the simulation can be seen in the Input Voltage to Motor figure. By comparing the simulation results with the real-world outcomes, the correctness of the theoretical helicopter model can be verified.

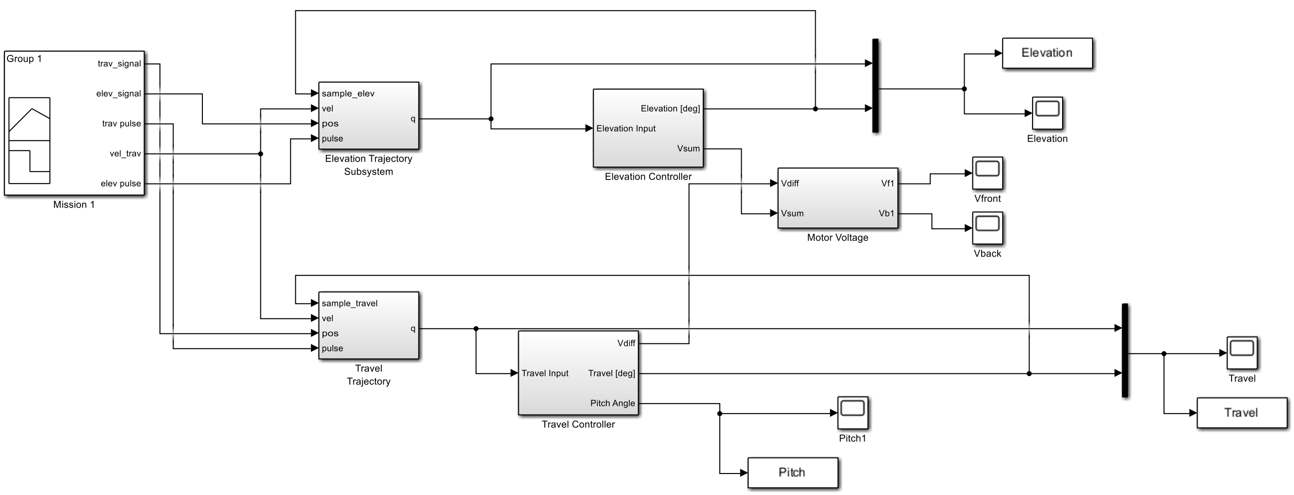

Figure 31: Simulation Model

Trajectory planning was used to track the performance of the model. Based on the simulation results, a smooth path plan was created where the helicopter constraints were met. As seen in the figure below, the voltage command for each motor was kept relatively low, not reaching the maximum voltage. These results are comparable to the maximum motor input voltage. Based on the system response, voltage fluctuation can be observed during the set-point changes. This model accounted for the amplifier gain by dividing the control input by the amplifier gain to estimate the voltage command for the front and back motors.

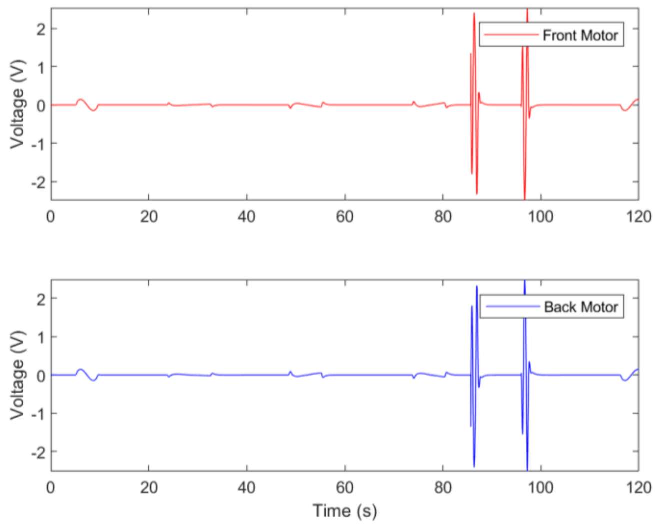

Figure 32: Input Voltage to Motor

The commanded voltage during pitch in the 3-DOF helicopter controller indicates the amount of electrical power supplied to the motors responsible for generating the necessary torque to pitch the helicopter. Before finalizing the system model, the pitch motion commanded higher allowable voltages resulting in higher torque generation and aggressive pitch motion. This led to the system being greatly unstable where further refinement of the simulation model and controller design was required. A voltage limit or saturation limit in the controller was decreased to prevent excessive voltage commands that could lead to hardware damage or instability when implemented in the real-world helicopter. Upper limits on the voltage commands were set based on motor constraints and safety margins to ensure that the performance remains within safe and stable operating limits. The voltage demand during pitch was kept relatively low once these changes were made. The helicopter model succeeded in pitch maneuvers based on the trajectory planner, without oversaturating.

The elevation, pitch and travel system response to the desired trajectory path are shown down below. The dynamics of the 3DOF helicopter model can be easily managed by the controllers developed in the Simulink model. The 3 DOF helicopter model responds as expected for the desired elevation and travel angle with fast settling times, performing each subsequent command swiftly.

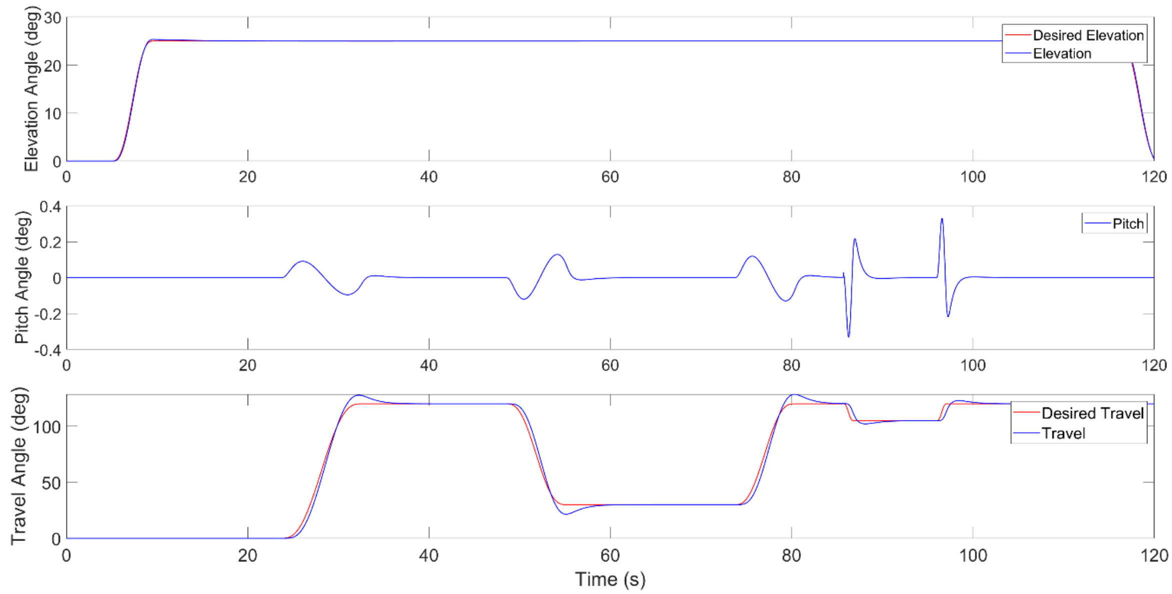

Figure 33: System Response to Trajectory Control

The travel trajectory simulation showed a promising trajectory path with a smooth transition to signal changes within the system. The controller design performed well based on the commanded trajectory path. The trajectory tracking performance reveals how well the control system design follows the desired trajectory. Here the tracking error, overshoot, settling time and steady-state error can easily assess the performance of the controller in terms of its accuracy and efficiency. The trainset response of the system in the simulation reveals some overshoot; a 10-degree deviation in the commanded angle of 120 degrees is observed. However, the system quickly recovers and reaches the desired trajectory within 5 seconds with smooth transitions between different states. This simulation result identified transient behaviour such as oscillation that may affect the performance and stability of the helicopter once deployed.

Based on the elevation simulation, the controller does an excellent job of following the elevation trajectory path with little to no deviation from the commanded angles.

Overall, the simulation results of the controllers using trajectory planning provide comprehensive insights into the model's accuracy, performance, stability, robustness, and sensitivity to signal changes. Analyzing these results allowed the team to refine the controller design, optimize parameters, and ensure reliable trajectory tracking in the real-world helicopter.

9.2 CONTROLLER IMPLEMENTATION

Based on the PID controllers and trajectory planning developed for the helicopter, the gain values were updated via manual tunning to reflect better stability within the real -world helicopter. Manual tunning was done by analyzing the mission one sequence and adjusting the gains accordingly. The final controller gains are outlined below in table, these values will now be used to conduct real-world test and evaluate the performance based on the mission profile outlined in the project statement.

Table 3: PID controller values

	Kp	Ki	Kd
Elevation	15	11	20
Pitch	3.5	0	2.5
Travel	1	0	1.5

9.3 TRAJECTORY RESULTS

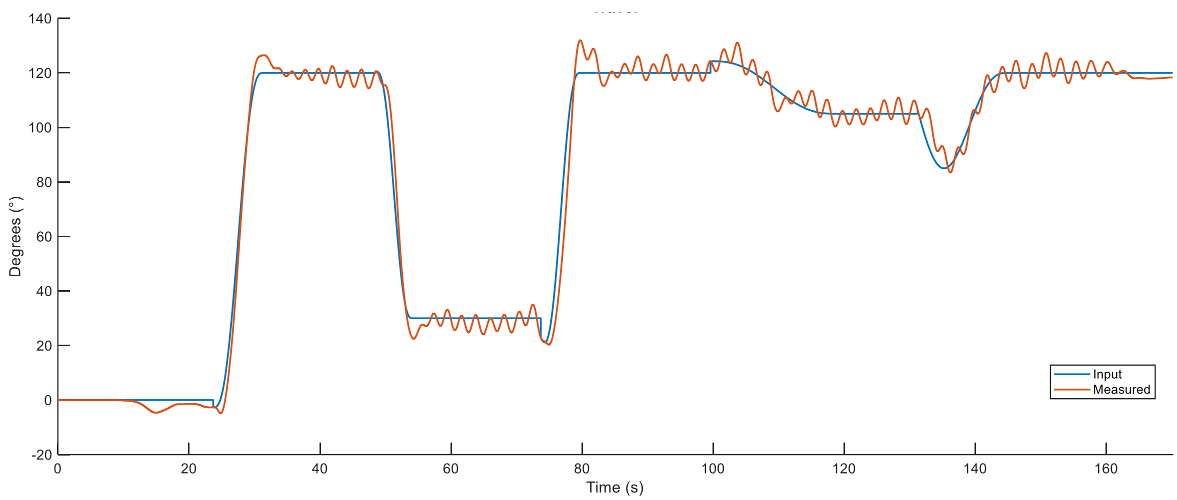

Figure 34: Mission 1 Travel Results

The results obtained from part one of the mission profile testing can be seen above. Here, it is evident that oscillation is present during steady-state operations. During the mission profile, the helicopter is required to stabilize at SLF at a travel and elevation positioning. However, the final system responds closely to the trajectory path once a commanded elevation and travel angle is sent to the controller. This suggests an instability issue specific to certain operating conditions of the helicopter and require further analysis to address the problem. Based on the controller implemented on the helicopter the controller gain may be inadequate to perform the maneuvers of mission one and two. The proportional gain of the travel controller can easily cause oscillation in a system if the gain is too high. A similar issue with the derivative gain can also arise. Although the derivative gain contributes to damping effects, too high of a gain can led to instability and overshoot in the system. The controller requires further improvement and thorough testing under various operating conditions to ensure a stable and controlled response from the helicopter. Fine-tuning of the travel controller ensures that the system and controller perform.

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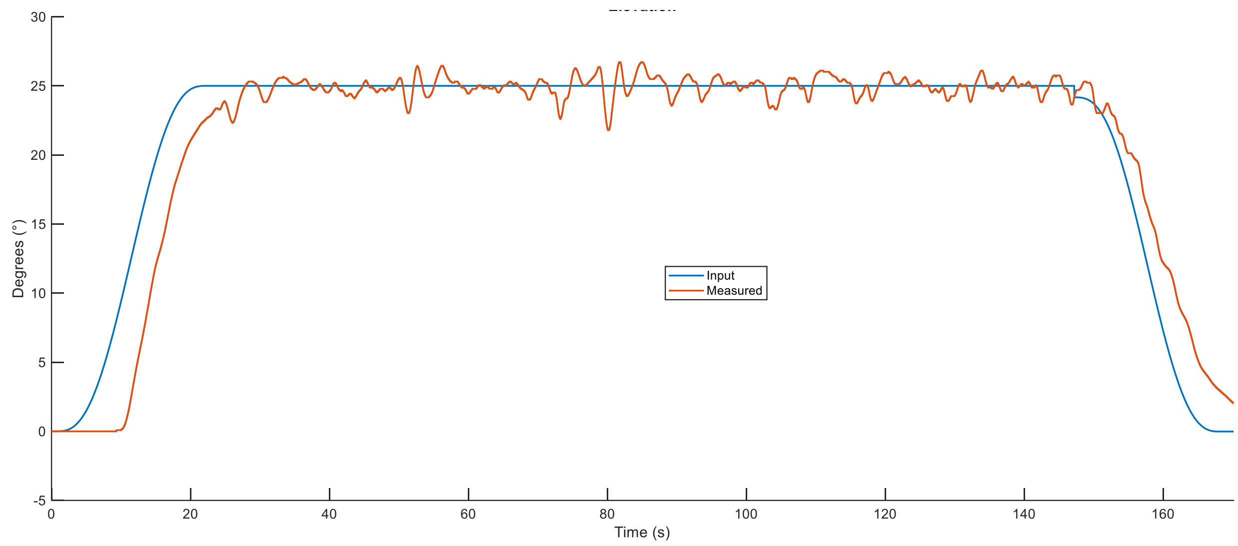

Figure 35: Mission 1 Elevation Results

The elevation result obtained from Mission 1 is provided above. The response of the measured elevation data during the take off and landing stage is time-delayed compared to the desired elevation. The time delay caused by many factors such as the physical distance between the station and the helicopter, processing time, and the others that affect that transmission of information within the systems. During the steady level flight, the measured elevation data is oscillating constantly. The oscillation commonly caused by the high Kp value. Regardless of the oscillation and time-delay, the helicopter has followed the trajectory most of the time. However, for the better performance in the helicopter trajectory, a better PID values should be provided to stable the helicopter and better communication environment to achieve the quick response of the helicopter.

10. HUMAN-MACHINE INTERFACE

10.1 INTRODUCTION TO HUMAN MACHINE INTERFACE (HMI)

Human-machine interfaces allow interaction between a user and the machine. One of the most common types of Human Machine Interface is Graphical User Interface (GUI). The most common example of this is the GUI used in cell phones [5]. Oftentimes designing such interfaces is challenging since it requires rigorous enhancements to be made user-friendly, accessible, and logical. There is a great deal of psychological aspects that affect the design of a GUI and it will be discussed in the following sections.

A system is developed with its complexities and intricacies that allows a machine to perform niche tasks. The complex mechanism and software that is used to enable the machine to perform is often not understood by viewers, others and oftentimes - even users. To resolve this problem, a human interface is highly recommended to be developed.

A human interface bridges the gap between humans and technology, enabling seamless interaction, enhancing usability, promoting accessibility, and ultimately enriching the user experience. By prioritizing the needs and preferences of users, well-designed interfaces contribute to the success and adoption of technology across various domains. The most prominent advantage of implementing a human interface is that the user does not require to understand the process or limitations of the machine to be able to operate it. The process and limitations of the system are considered during the design process and the human interface automatically limits the operation of the machine within these parameters.

ISA101 is an established committee that monitors the standards and practices for computer-based HMI [6]. The ISA101 outlines a lifecycle for the HMI from its inception to its operation and refining stage. The following figure outlines the stages of the HMI development.

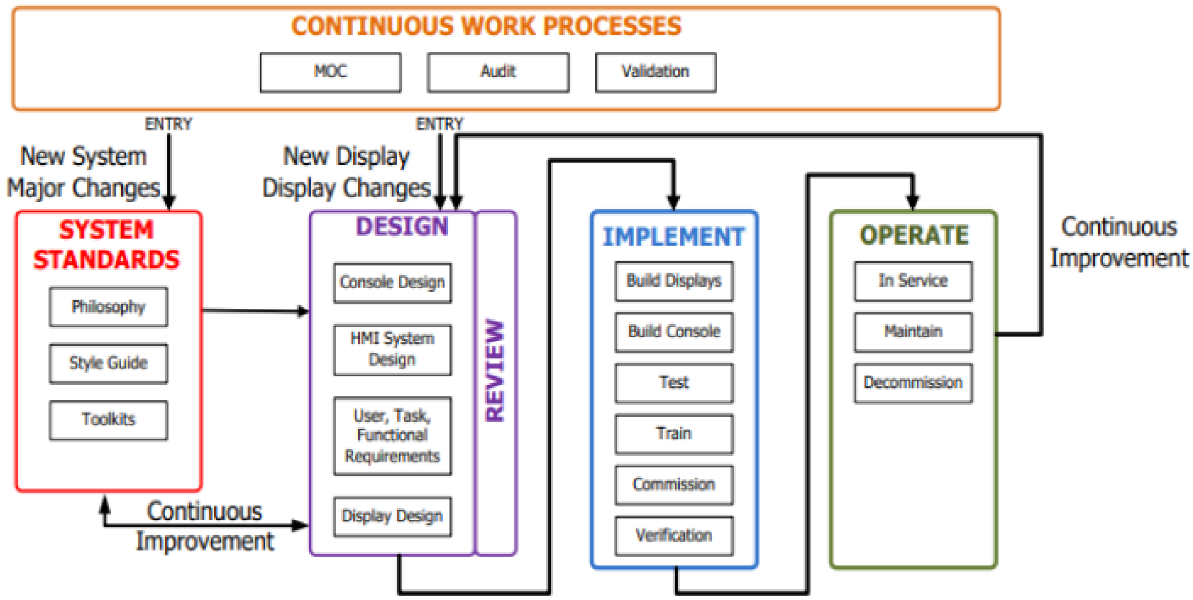

Figure 36: HMI Lifecycle

The above figure can be interpreted as continuous improvement throughout the process of developing HMI is important. A streamlined approach is shown in the above HMI development process. The tasks undertaken include system definition, system analysis, system design and Implementation and validation and verification.

10.2 INTRODUCTION TO GRAPHICAL USER INTERFACE (GUI)

One of the most common occurring types of human interface is the graphical user interface (GUI). Graphical User Interface (GUI) is a type of interface that allows users to interact with electronic devices through graphical icons and visual indicators, as opposed to text-based interfaces, typed command labels, or text navigation. GUIs have revolutionized the way users interact with computers, making them more intuitive and user-friendly. The following list details some of the features of a typical GUI.

1. Windows: A GUI typically consists of windows. These windows can be moved, resized, minimized, and closed.
2. Icons: Icons are graphical representations of files, folders, applications, or actions. Users can interact with icons by clicking on them to perform various tasks.
3. Menus: Menus provide a hierarchical structure of options and commands that users can access by clicking on menu titles. Common menu types include file menus, edit menus, view menus, and help menus.

4. Buttons: Buttons are graphical elements that users can click on to trigger actions or commands. They are often labelled with text or icons to indicate their function.
5. Dialog Boxes: Dialog boxes are windows that appear in response to user actions, such as opening a file or saving changes. They typically contain fields for user input and buttons for confirming or cancelling the action.
6. Scroll Bars: Scroll bars are graphical elements that allow users to scroll through content that does not fit entirely within a window or viewport. They can be vertical or horizontal and enable users to navigate through large amounts of information.
7. Graphical Feedback: GUIs provide visual feedback to users to indicate the outcome of their actions. This feedback may include changes in the appearance of icons or buttons, animation effects, or status messages displayed on the screen.

The designing of the PID controller and the development of the software to operate and execute specific tasks was done using the MATLAB software. The user interface was also developed using MATLABs app development tool - GUIDE. This was done because of straightforward communication between the Simulink models and the GUI. GUIDE was also used to allow other sub-teams to understand and provide constructive feedback on how to enhance the user experience and how to further increase the progress of the report at a better pace since the software to control the Quansor helicopter was being developed in MATLAB.

10.2.1 Requirements for GUI

A structured approach is used to manage the development, deployment, operation and disposal of the GUI and a systems engineering method is implemented.

10.2.2 System Definition

There are two missions that are to be performed with the Quanser helicopter. The first mission is a planned trajectory that the helicopter will follow at steady level-flight. The second mission is to approach a foam board and detect a red marker and drill a hole using an electric drill. The user of this system will only perform these two tasks with GUI. Therefore, the GUI shall have two buttons to control the flight operation.

10.2.3 System Analysis

A control panel is to be added in the GUI where the user can easily locate the buttons to push to initiate flight. A START button is used to bring the helicopter in a flight-ready state and to allow it to perform the missions as soon as the command is given to them. A STOP is to be added for emergency termination of the flight in case of any unforeseen circumstances or when flight is not necessary. If the termination of the flight is not necessary but the user wishes to land to analyze the flight data, a LAND button shall be implemented. A button to activate and to stop the drill is also implemented. A window should show the state of the drill when requested by the user. In flight mode, the user can observe the performance of the flight graphically. This is done by adding labelled graphs in the GUI. The interface on the screen needs to show where the helicopter is right now and where we want it to go, all in real-time. This way, we can make quick adjustments and give the helicopter the right instructions as it moves.

10.2.4 System Design

The goal while developing the GUI is to allow a human being logical and sentient enough to use a smartphone. Since the 3DOF helicopter is controlled with Simulink, the GUI is also developed using MATLAB software and tools to allow easier interchangeability.

The GUI screen will have an output that allows the machine to keep the user updated on the progress of the flight mission. The outputs include a graph of the state of the elevation, travel, and pitch. The intent of this is to allow the user to examine the stability of the mission. On the same screen as the GUI, the output information of the machine will be displayed. The information that shall be displayed on the GUI screen includes, but is not limited, to the following.

- Current Mode of the 3DOF helicopter (Running/Idle)
- Current state of Mission (Mission 1/ Mission 2)
- Current state of Drill (Running/ Idle)
- Current state of Landing (Landing Mode)

The above states mentioned can be displayed by text beside the push buttons of the functions respectively. A new GUI model was created in MATLAB and an empty app layout was created as shown below.

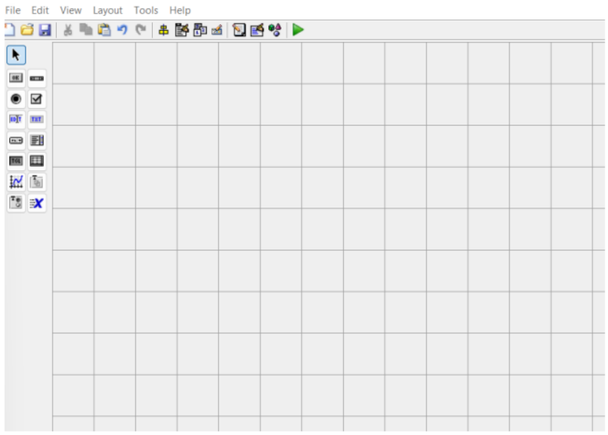

Figure 37: Building GUI in GUIDE

To switch between the states of flight, index vectors are used. If the constant1 block has a value of 0, the first circuit line is operated, if the value of constant1 block is 1, the next circuit line is operated. Index vectors allowed multiple operations to be performed by the same circuit design and automated the process. This automation led to a robust system for the GUI development.

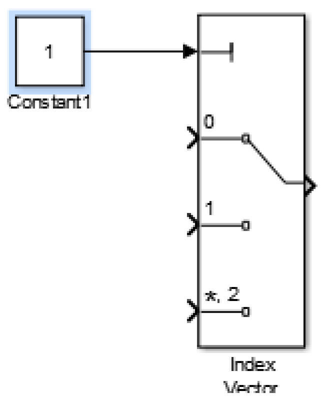

Figure 38: index Vector Simulink Block

Simulink Control Panel: The GUI is designed in such a way that any other file is not needed to be opened by the user. The user is required to open the app. The program file will be opened in the background when the user clicks LOAD. The user is not required to know the name of the program file. As the user clicks on the LOAD button, an open function is used to open then file and the BUILD button uses the 'qc_build_model' function to build the Simulink file.

Once the program file is opened in the background, the user observes this action and proceeds to build the Simulink model. The need to access the Simulink file and then build it from the toolbars in the program file is eliminated, thus making it quicker, efficient and more user friendly.

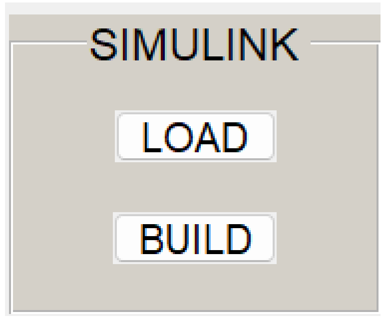

Figure 39: Simulink Control Panel

Initialize Control Panel: A green button is used to identify the button to start the program. This button runs the program but does not allow the helicopter to enter flight mode. A STOP button is used to terminate the Simulink model. Green is considered as a 'Go' command and red is often contextualized with 'Stop'. The need for the STOP button to be in bright red and bigger is necessary, this allows the user to easily locate it to terminate the program in an event of close contact with surroundings or when the flight continuation becomes a safety risk.

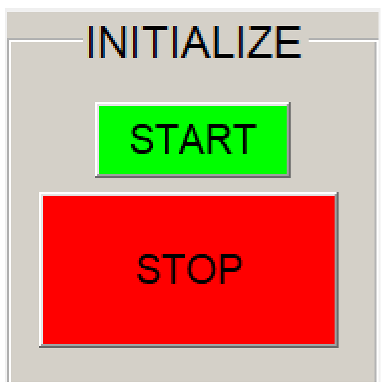

Figure 40: Initialize Control Panel

Drill Control panel: The drill is communicated through a Bluetooth receiver and transmitter module. There are 5 states to the drill, but the GUI user needs to control only two states, namely - Active state and Stop state. The Activate button turns the drill on and intuitively, the Stop Drill button turns it off. The Drill Control panel incorporates a button to show the state of the drill. This is a very important feature since the user might forget if the drill is on or off. To determine the state of the drill, the user is required to push the Drill Sate button and the text window on the right will update to show the state number.

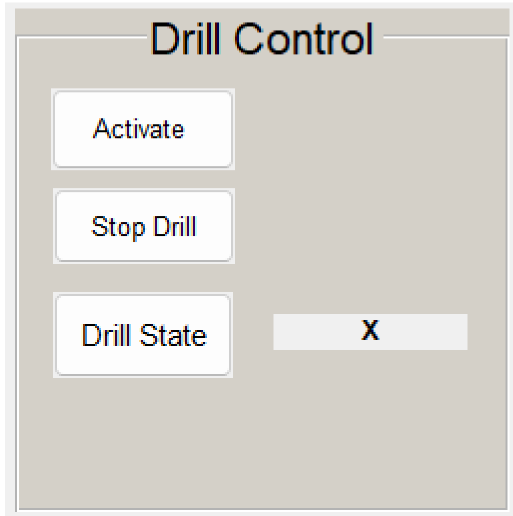

Figure 41: Drill Control Panel

Mission Control Panel: This panel is analyzed to be the most used panel. Through that panel, the user will control the flight mode of the drone. Soon after the User presses MISSION 1, the drone will take off and come to steady-level flight and then follow a trajectory path and eventually come land. As the user presses MISSION 2, the helicopter takes off and approaches a foam board to detect it. As soon as the board is detected, the drill is turned on automatically and does not require the user to turn it on from the Drill Control Panel.

Figure 42: Mission Control Panel

Real-Time Feedback: When the helicopter enters flight-mode, the user can observe the performance of the flight path. Three plots are displayed on the GUI and they are linked to show the desired vs actual response for Elevation, Pitch and Travel respectively. The plots are labeled so the user can identify them and the axises automatically update to portray the current flight path.

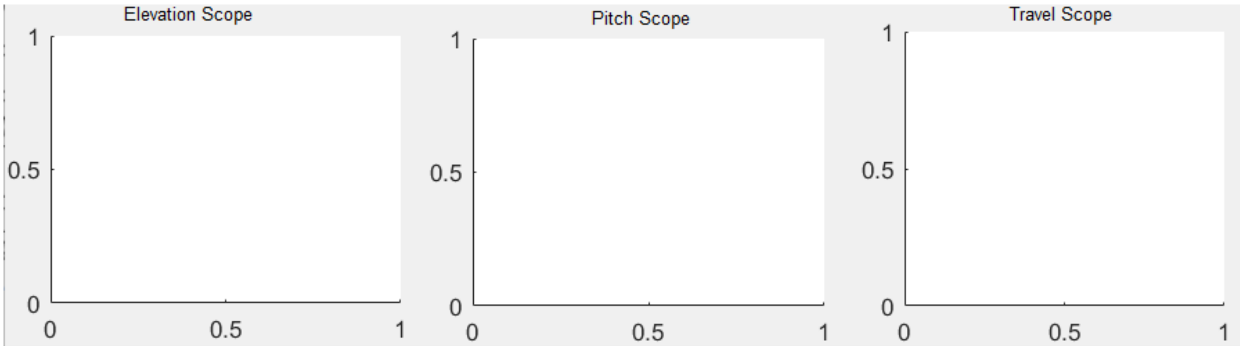

Figure 43: Real-Time Feedback

Panel Placements: The panels discussed above are strategically laid out on the GUI. The Simulink panel is located above the Initialize control panel. This will seem intuitive to the user, and they will know to load and build the model before pressing start. The STOP button is enlarged on the bottom left corner. This allows the user to locate it easily under any circumstances. The drill control panel is located on the bottom center of the GUI window. The state of the drill is important for the user to know so that the user does not approach the drill while it is in active state. The mission control panel is located on the bottom right window of the GUI. After the user has started the program and built it and observed the drill state, they can proceed to initiate missions as required. Commanding to start a mission is the last command the user inputs, therefore the mission control panel was situated on the bottom right corner. The scopes for travel, elevation and pitch are located on the top to enable the user to observe the performance of the Quanser helicopter easily.

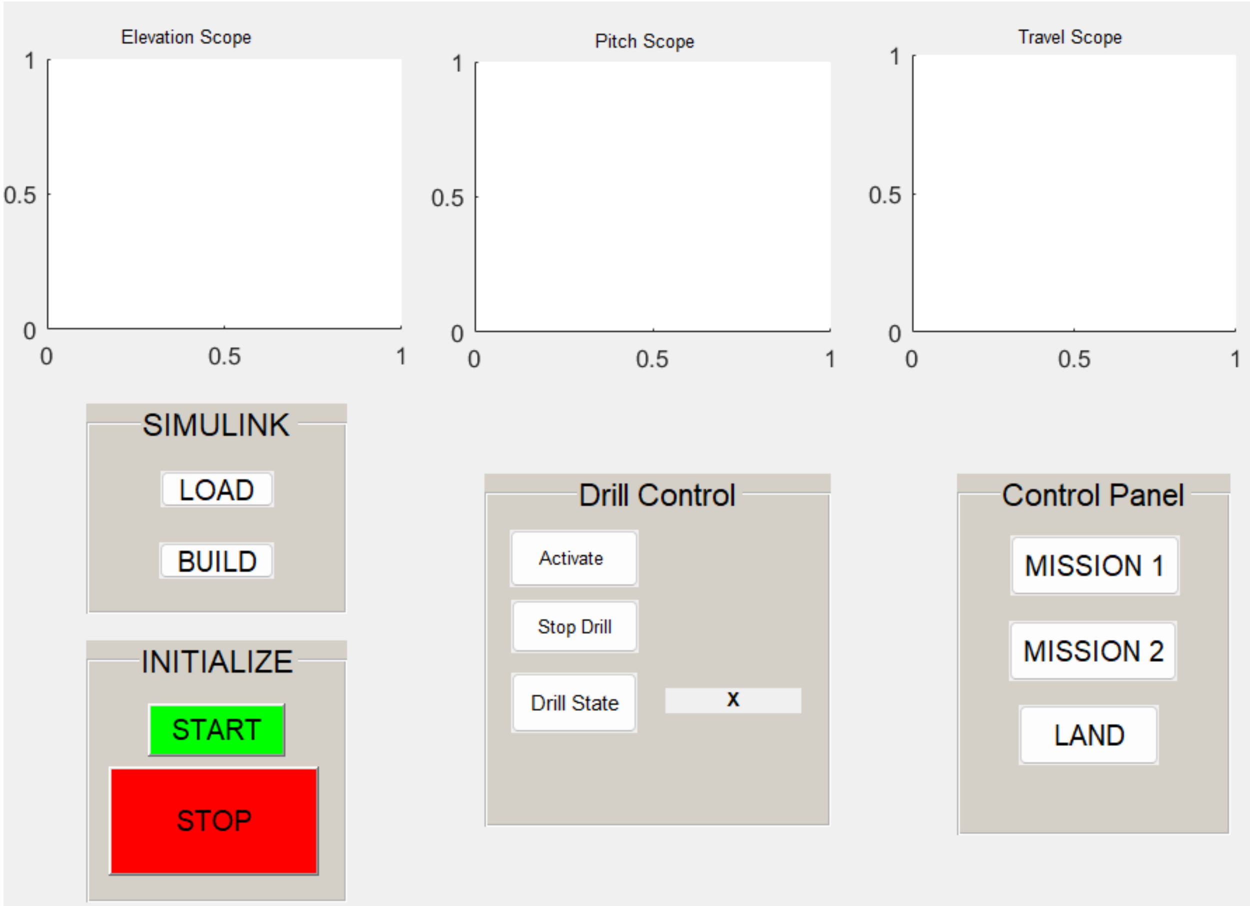

Figure 44: GUI

10.2.5 Implementation

The Code for the GUI was streamlined better, and the location of the panels and font size of the texts was altered to be more user-friendly. The system was verified and validated that it meets the needs and requirements of the stakeholders and that the GUI operates in its intended environment. All the buttons of the GUI performed their intended functions. The GUI functionality was tested repeatedly in all possible scenarios, and it did not fail. The verification process of the GUI was performed with attention to detail and keeping system requirements at priority.

11. CONCLUSION

In summary, this project aimed to simulate and control the Quasar 3-DOF Helicopter through control system design, trajectory control, signal processing and a human-machine interface.

A theoretical study of the 3 DOF helicopter using control system theory and design was conducted before implementing control of the helicopter in the real world. The dynamics of the 3DOF helicopter were modelled and simulated to evaluate the controller design before implementation. The 3 DOF helicopter model responded as expected for the desired elevation and travel angle with fast settling times, performing each subsequent command swiftly. These results were then implemented in a real-world helicopter to evaluate its performance and trajectory control.

During this process, signal processing was also implemented for visualization, sensor-computer optimization, and trajectory control. Converting processing components such as transfer functions into Bode plots facilitates visualization and analysis of system features, enabling optimization through fine-tuning. Fine-tuning in trajectory control aims to generate smoother motion, and reduce overshoot, settling time, and steady-state error.

Furthermore, with the interactive use of GUI, the control of the helicopter was enhanced to follow the trajectory. An in-depth understanding of the user interface was applied to the approach from the perspective of a first-time user. Also, the real-time axis of the helicopter in pitch, elevation, and travel was presented in real-time with the input and actual data to show the user how well the helicopter is performing the mission profiles.

The combination of many approaches; System Engineering Process, System Modeling, Control Design, Trajectory Planning, Signal Processing, and Human Machine Interface were applied to optimize the mission profile and achieve the mission profile successfully.

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